

Lesson 02 Demo 02

Building a Research Agent Workflow with MCP Server in n8n

Objective: To build an agent workflow in n8n that integrates with the MCP server by using Docker, a chat trigger, Google Gemini, memory, and SerpAPI.

Tools required: None

Prerequisites: None

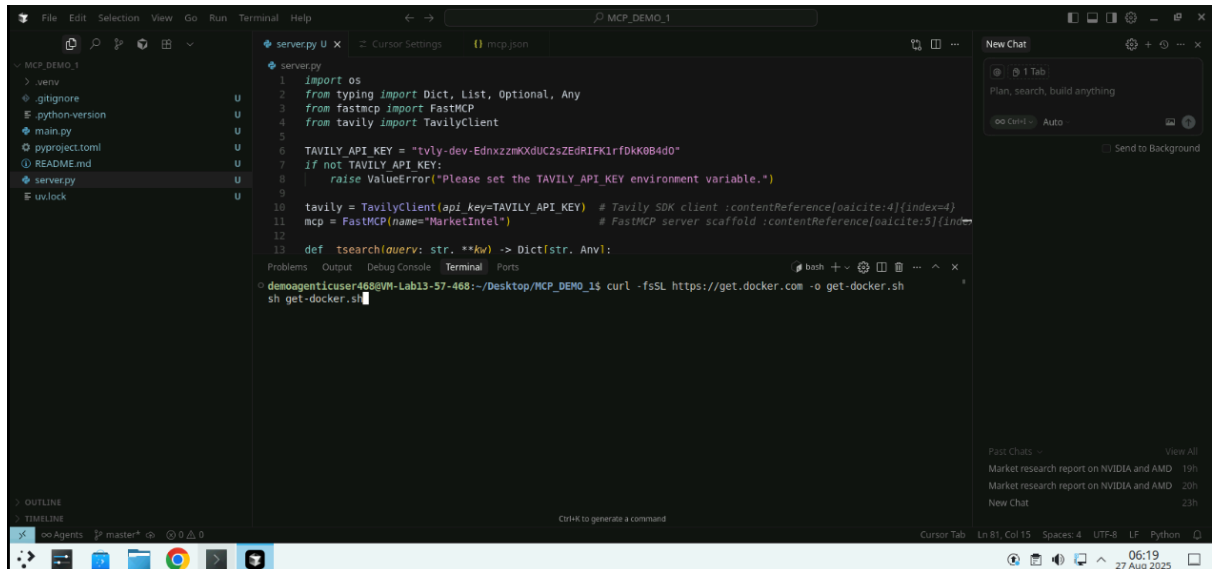
Steps to be followed:

1. Install n8n using Docker
2. Add the chat trigger and AI agent
3. Integrate the Google Gemini LLM chat model
4. Add memory and MCP Client tool
5. Create an MCP server trigger in n8n
6. Integrate the SerpApi
7. Add the SerpApi key and activate the server trigger
8. Run the Research Agent

Step 1: Install n8n using Docker

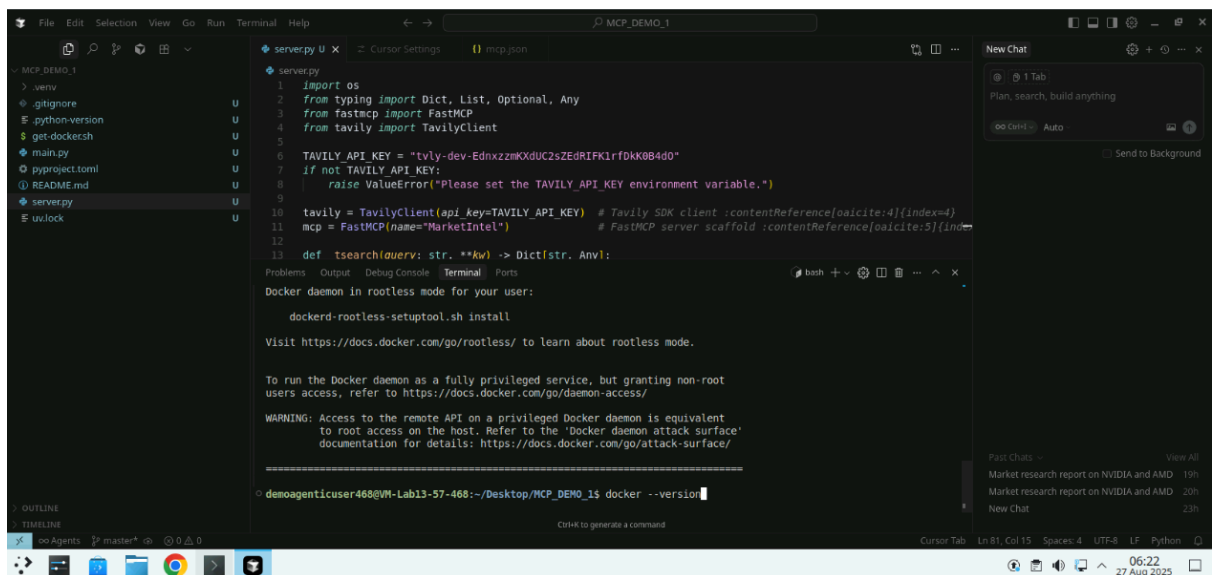
1.1 Open the Cursor AI terminal and paste the following Docker installation command:

```
curl -fsSL https://get.docker.com -o get-docker.sh
sh get-docker.sh
```

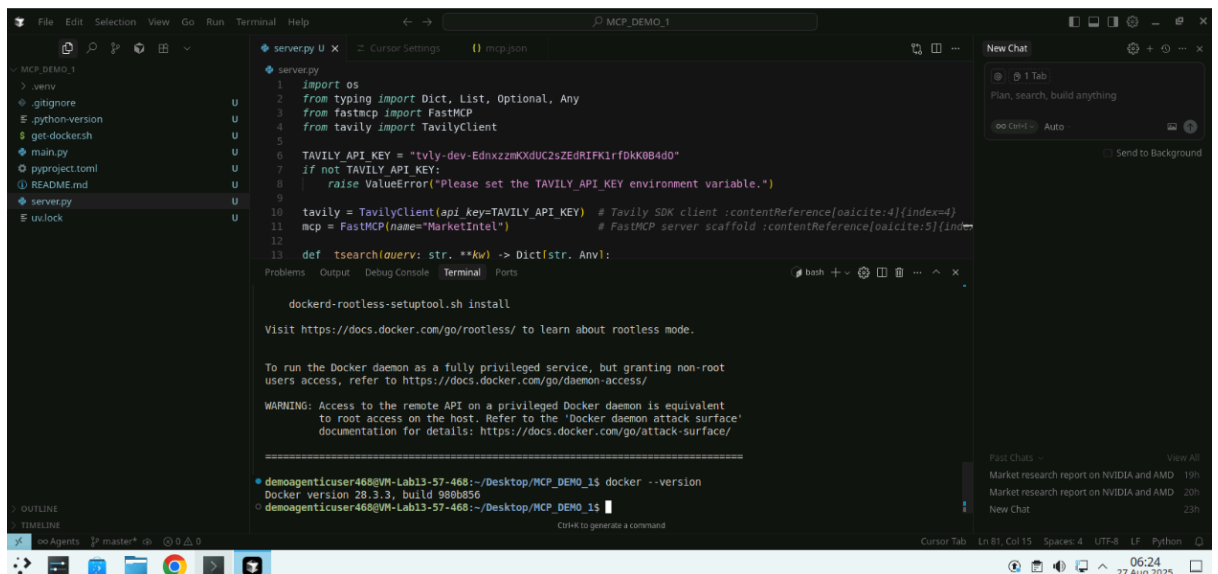


1.2 After installation, verify the installed version by running the following version check command:

```
docker --version
```



The Docker version will be displayed



The screenshot shows a VS Code editor with a file explorer on the left, a Python script named `server.py` in the center, and a terminal window at the bottom. The script defines a FastMCP server named "MarketIntel" using the Tavily API. The terminal window shows the output of the `dockerd-rootless-setup` command, including instructions on how to run Docker as a non-root user and a warning about access to the remote API. The terminal also shows the output of the `docker --version` command, which displays "Docker version 28.3.3, build 988b856".

```
1 import os
2 from typing import Dict, List, Optional, Any
3 from fastmcp import FastMCP
4 from tavily import TavilyClient
5
6 TAVILY_API_KEY = "tvlv-dev-EdnxzzmKXduC2sZEdRfK1rfDKK084d0"
7 if not TAVILY_API_KEY:
8     raise ValueError("Please set the TAVILY_API_KEY environment variable.")
9
10 tavily = TavilyClient(api_key=TAVILY_API_KEY) # Tavily SDK client :contentReference[oaicite:4]{index=4}
11 mcp = FastMCP(name="MarketIntel") # FastMCP server scaffold :contentReference[oaicite:5]{index=5}
12
13 def tsearch(query: str, **kw) -> Dict[str, Any]:
14     """Search the web for the given query using Tavily.
15     Returns a dictionary with 'results' and 'raw_html' keys.
16     """
17     results = tavily.search(query=query, **kw)
18     return {"results": results.get("results", []), "raw_html": results.get("raw_html", {})}
19
20 if __name__ == "__main__":
21     mcp.add_tool(tsearch)
22     mcp.run()
```

```
dockerd-rootless-setup: sh install
Visit https://docs.docker.com/go/rootless/ to learn about rootless mode.

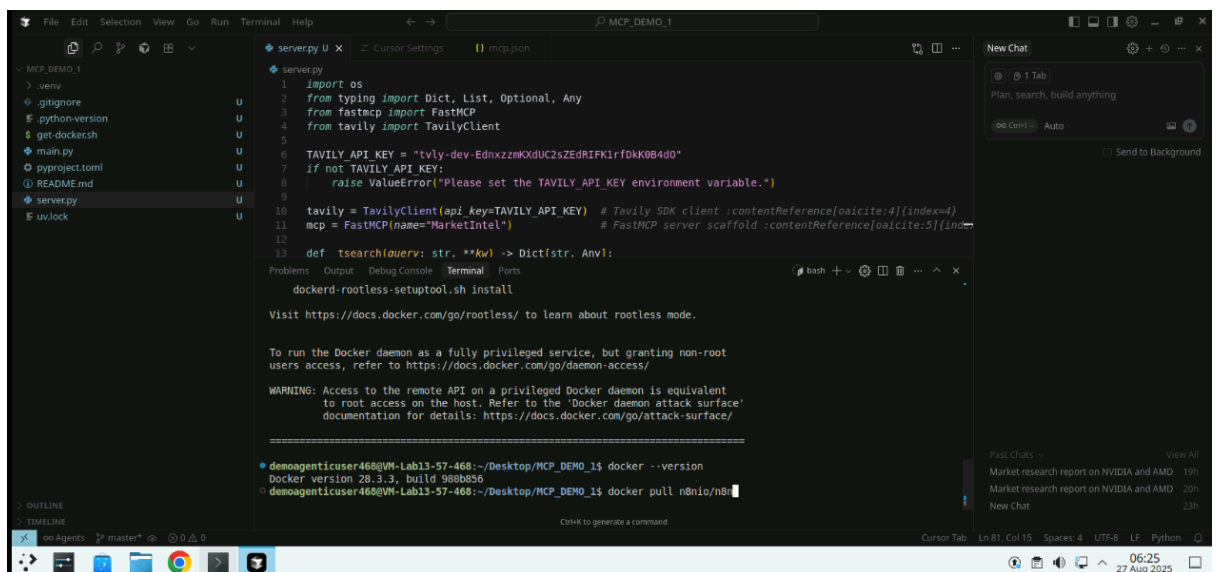
To run the Docker daemon as a fully privileged service, but granting non-root
users access, refer to https://docs.docker.com/go/daemon-access/

WARNING: Access to the remote API on a privileged Docker daemon is equivalent
to root access on the host. Refer to the 'Docker daemon attack surface'
documentation for details: https://docs.docker.com/go/attack-surface/

demoagentuser468@VM-Lab13-57-468:~/Desktop/MCP_DEMO_1$ docker --version
Docker version 28.3.3, build 988b856
demoagentuser468@VM-Lab13-57-468:~/Desktop/MCP_DEMO_1$
```

1.3 Install n8n locally by running the following command:

sudo docker pull n8nio/n8n



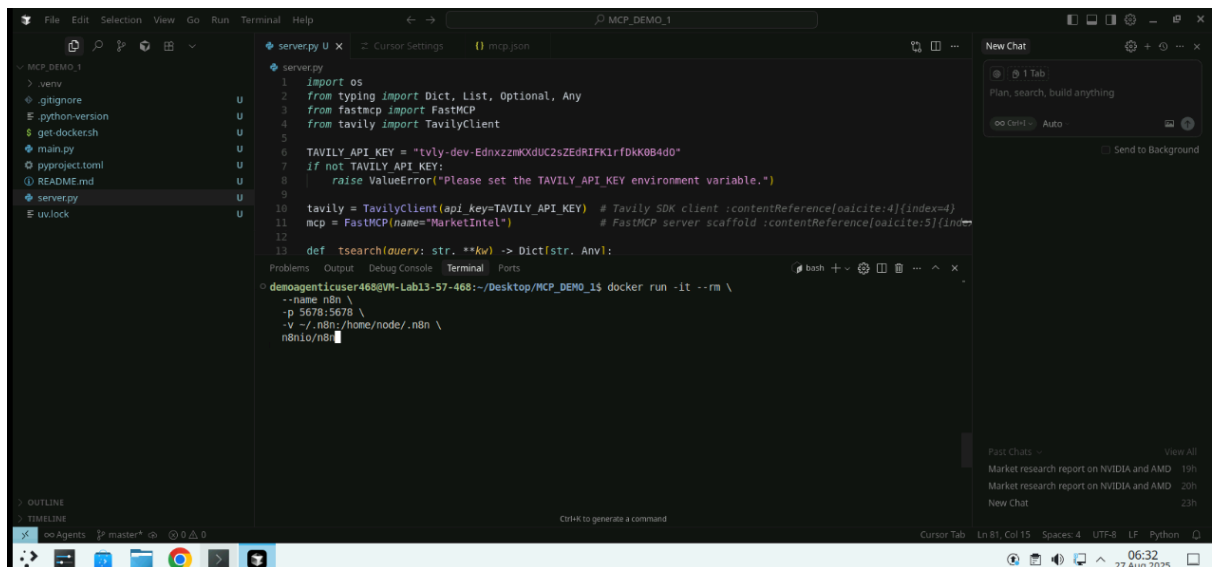
The screenshot shows the same VS Code editor as before, but the terminal window now displays the command `sudo docker pull n8nio/n8n` being executed. The command is partially typed, and the cursor is at the end of the line.

```
demoagentuser468@VM-Lab13-57-468:~/Desktop/MCP_DEMO_1$ sudo docker pull n8nio/n8n
```

1.4 Initialize an n8n workflow by entering the following initialization command:

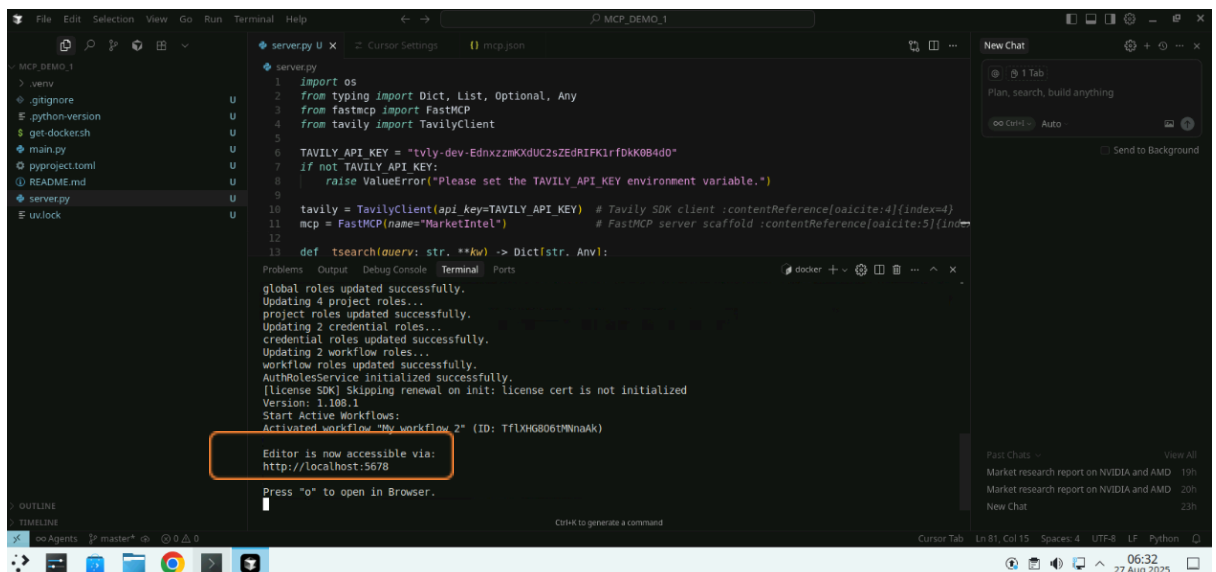
```
sudo docker run -it --rm --name n8n -p 5678:5678 -v ~/.n8n:/home/node/.n8n  
n8nio/n8n
```

```
sudo docker run -it --rm \ --name n8n \ -p 5678:5678 \ -v  
~/.n8n:/home/node/.n8n \ n8nio/n8n
```

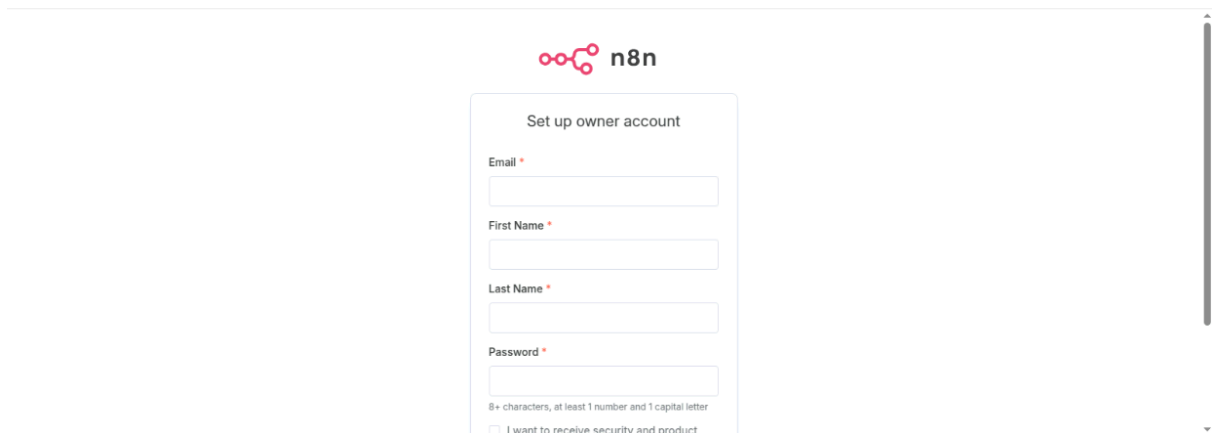


1.5 Copy the following link and paste it into your browser:

<http://localhost:5678>

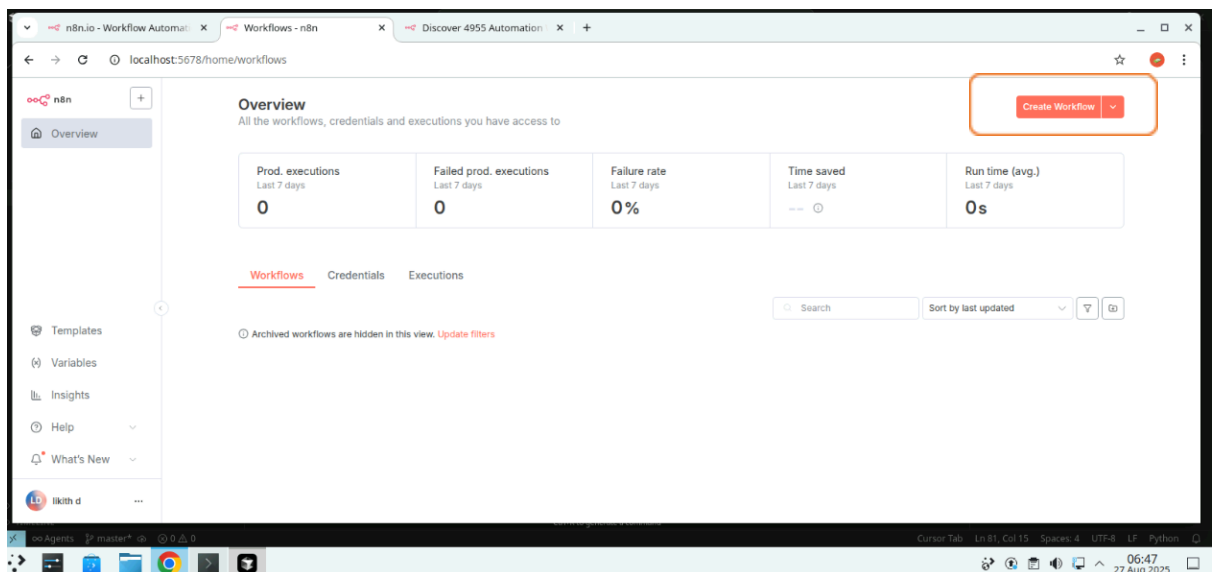


- 1.6 After pasting the link into your browser, you will see the login screen; sign in using your credentials.



The image shows the n8n login screen. At the top center is the n8n logo. Below it is a form titled "Set up owner account". The form contains four input fields: "Email", "First Name", "Last Name", and "Password". Below the "Password" field is a note: "8+ characters, at least 1 number and 1 capital letter". At the bottom of the form is a checkbox labeled "I want to receive security and product".

- 1.7 After login, the n8n dashboard will appear; click on the **Create Workflow** button in the top-right corner.

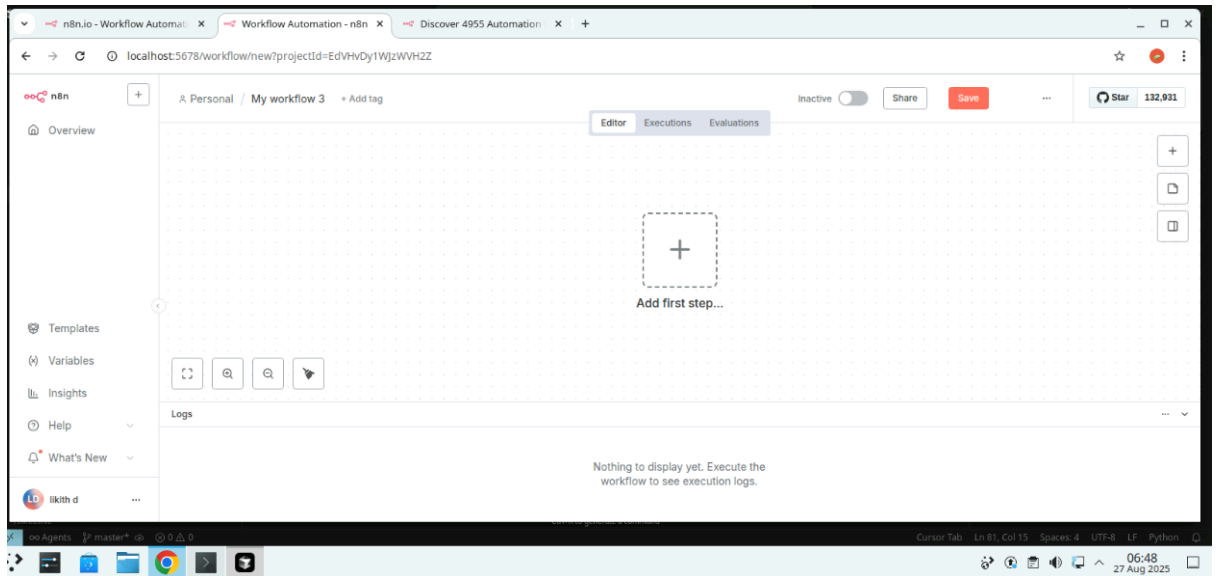


The image shows the n8n dashboard overview. The top navigation bar includes the n8n logo and a "+" button. The main content area is titled "Overview" and contains a table with the following data:

Prod. executions Last 7 days	Failed prod. executions Last 7 days	Failure rate Last 7 days	Time saved Last 7 days	Run time (avg.) Last 7 days
0	0	0%	--	0s

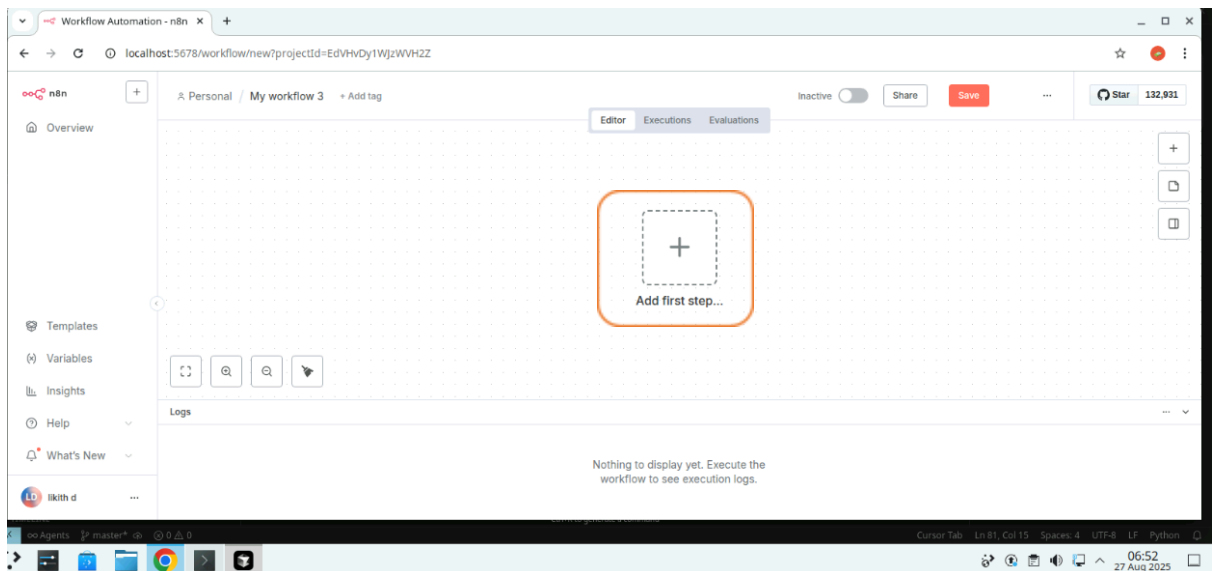
Below the table are tabs for "Workflows", "Credentials", and "Executions". The "Workflows" tab is selected. A search bar and a "Sort by last updated" dropdown are visible. A red box highlights the "Create Workflow" button in the top-right corner.

n8n is now initialized and running locally via Docker.

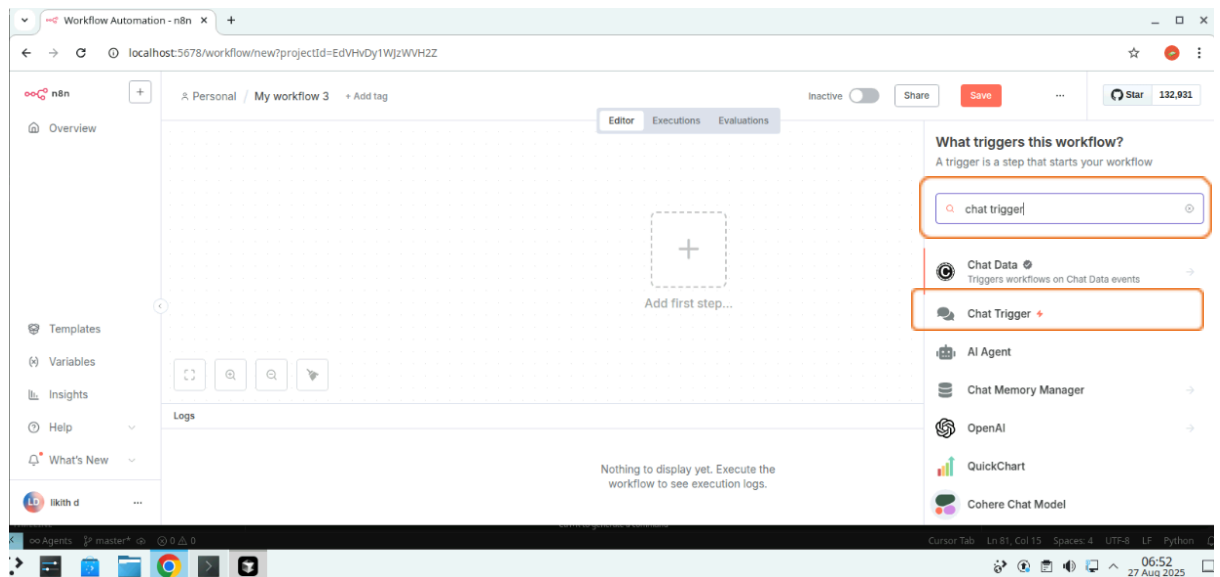


Step 2: Add the chat trigger and AI agent

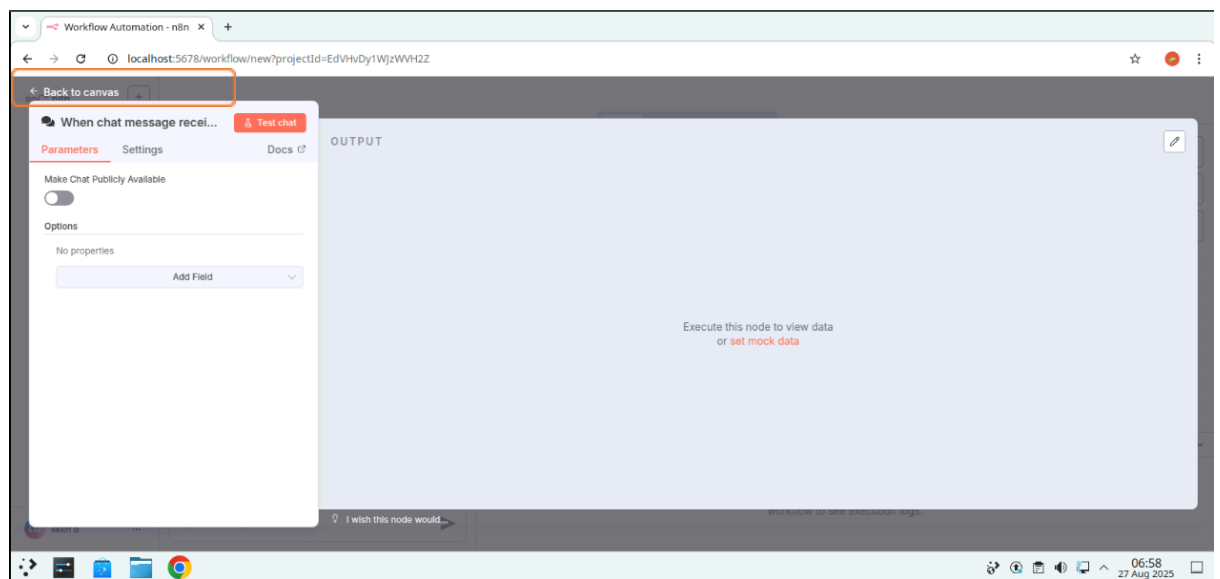
2.1 In the workflow editor, click on Add First Step



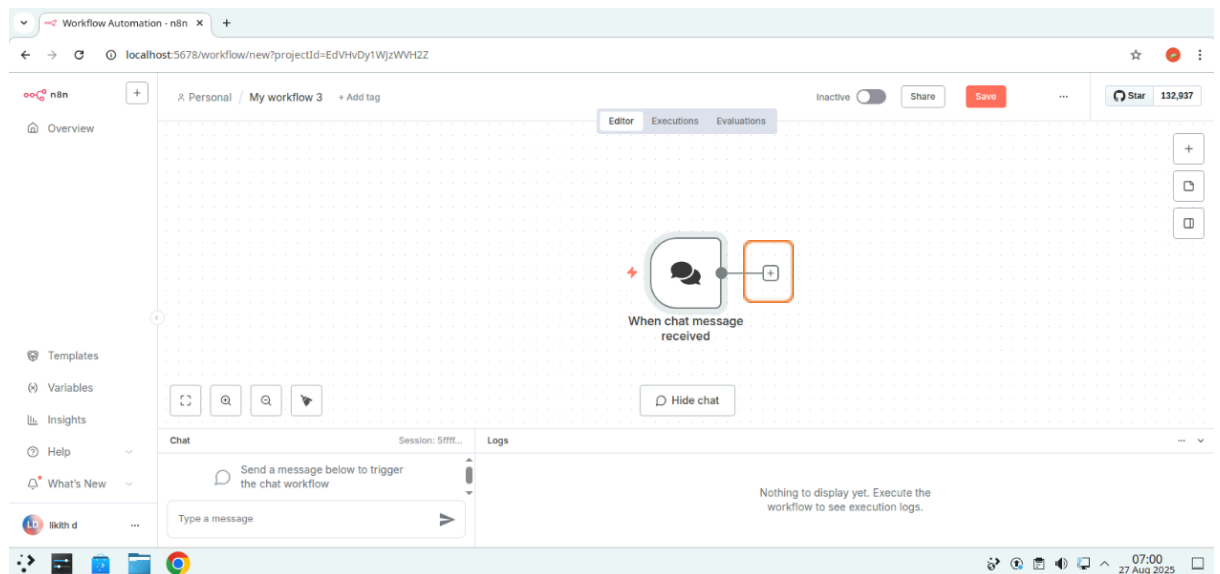
2.2 In the search bar, type **Chat Trigger** and select it



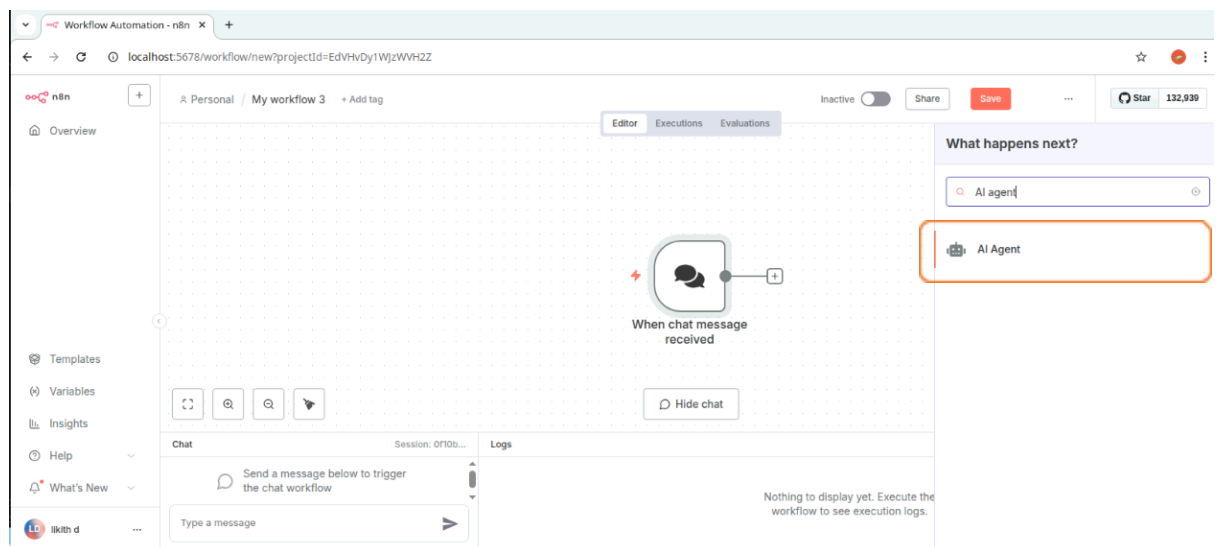
2.3 Click **Back to canvas** button on the top-left side once the trigger appears



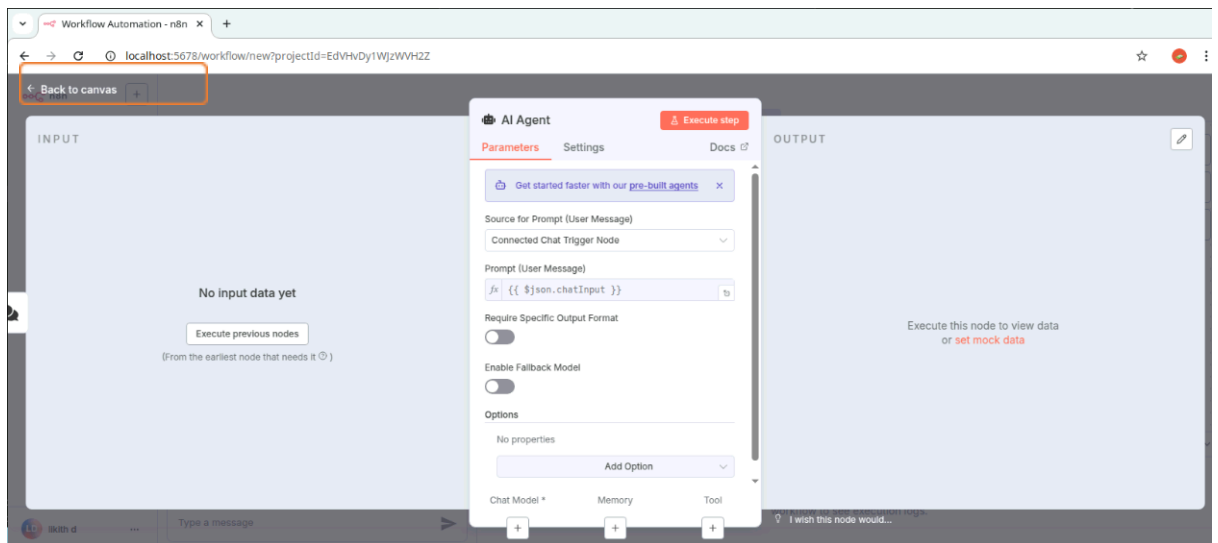
2.4 Add an agent node by clicking the + button



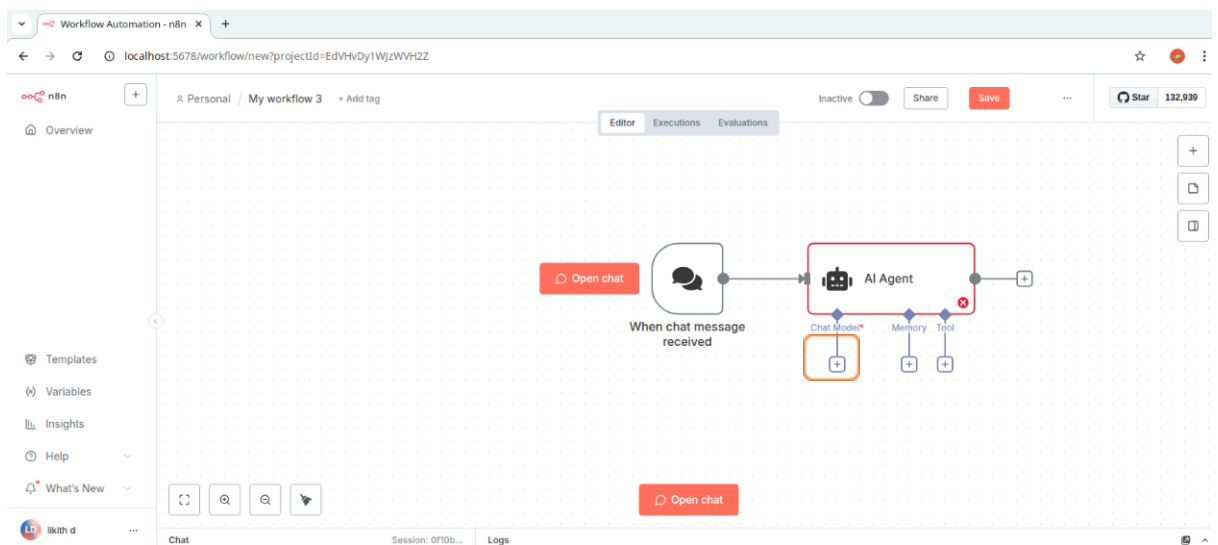
2.5 In the search window, search for the AI agent and select it



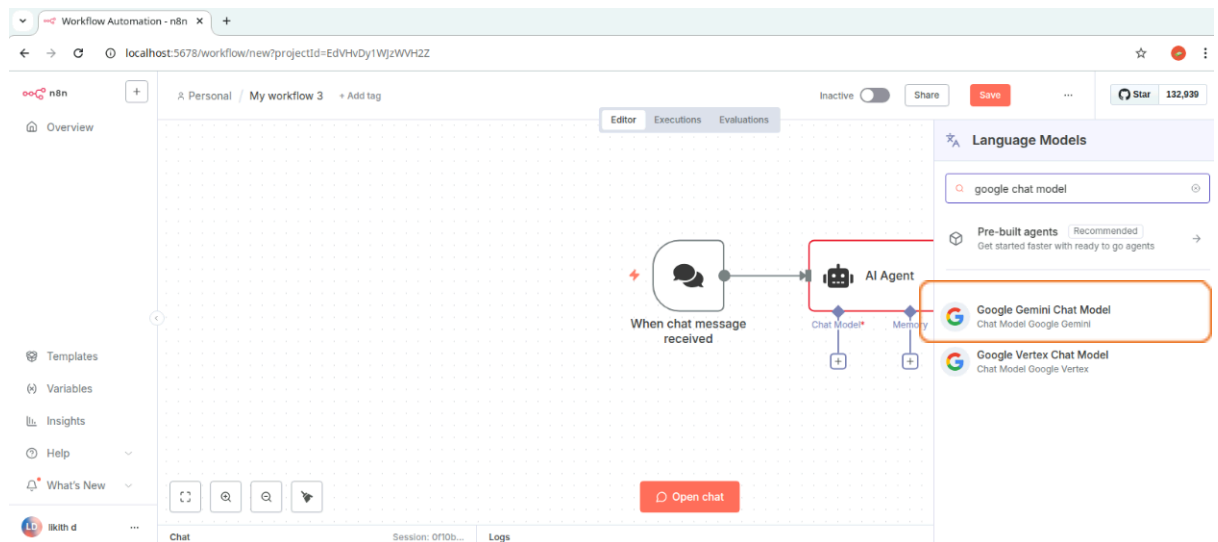
2.6 After the AI Agent window appears, click **Back to canvas**



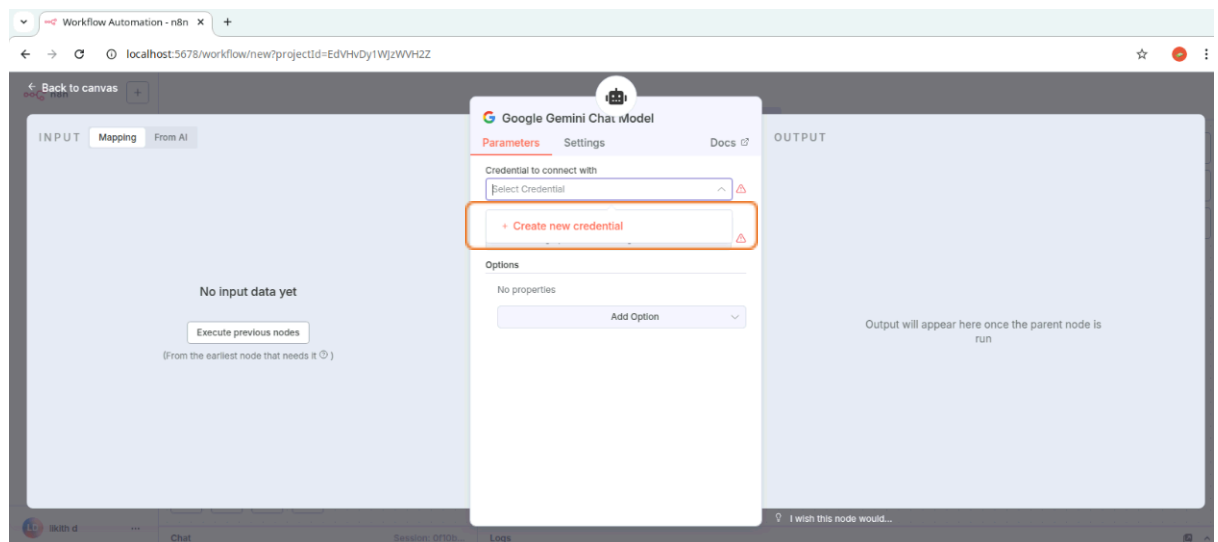
2.7 Click the + button on the Chat Model section to connect it to a language model



2.8 In the search bar, type **Google Chat Model** and select **Google Gemini Chat Model** from the LLM options

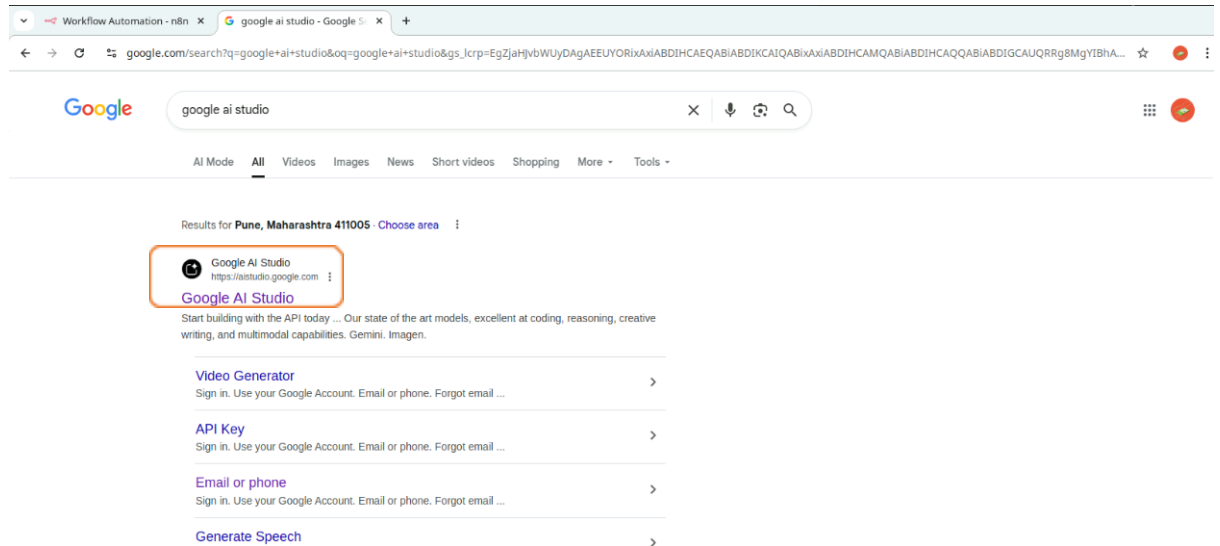


2.9 In the **Credentials** section, click **Create New Credential** to proceed with integration

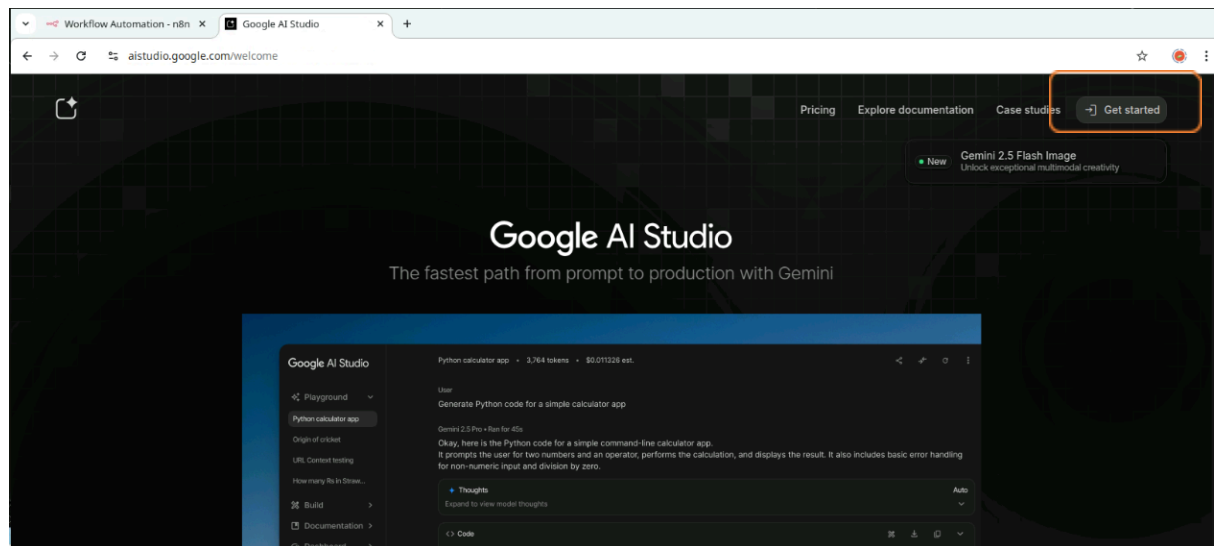


Step 3: Integrate the Google Gemini LLM chat model

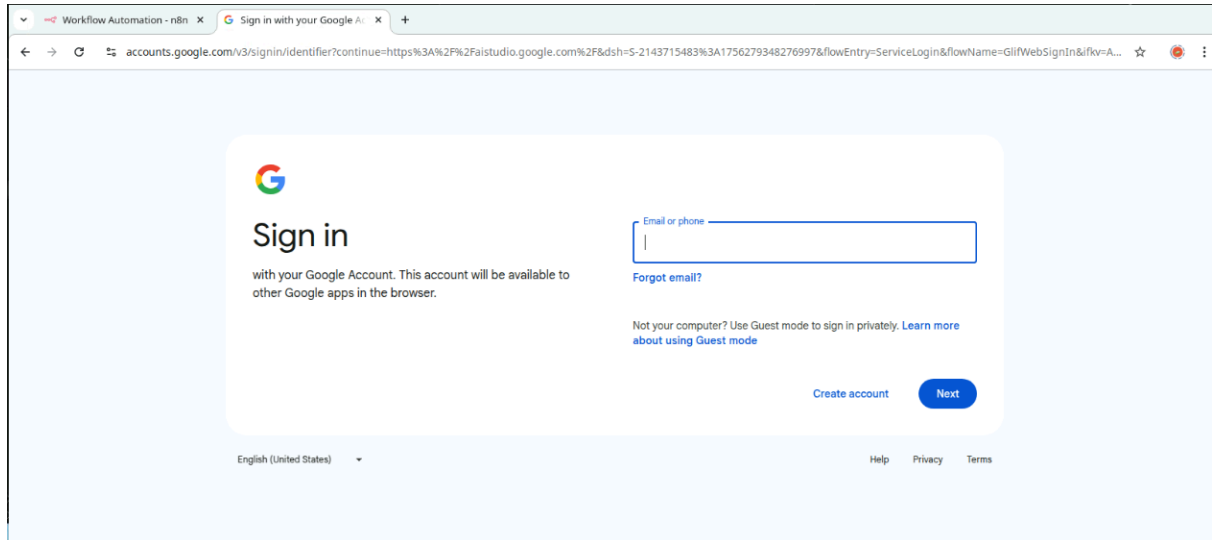
3.1 Open a new browser window, search for **Google AI Studio**, and open the official site



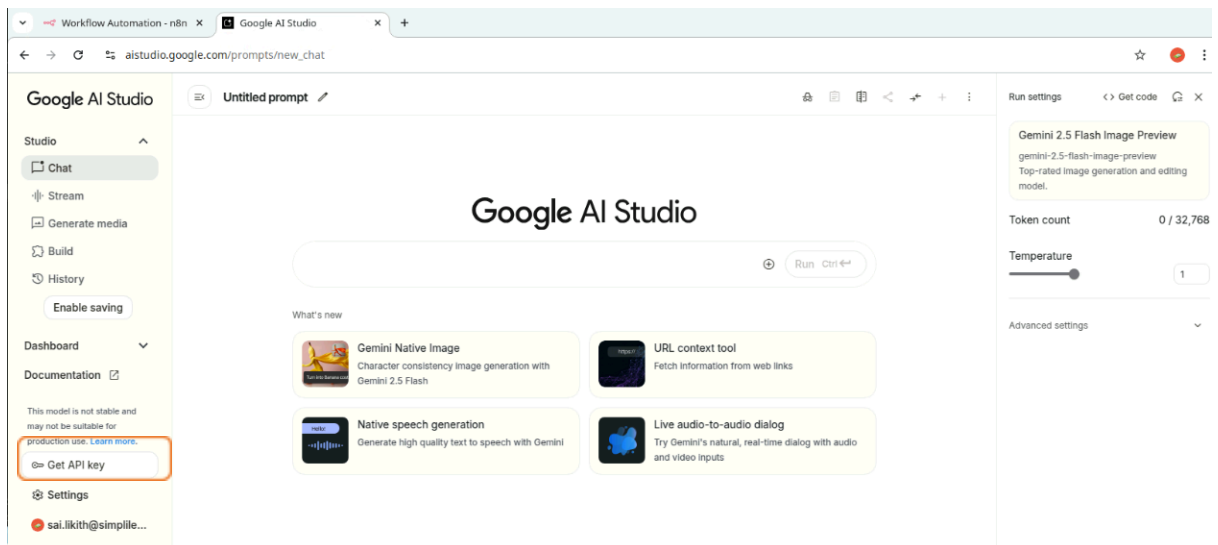
3.2 Click the **Get started** button in the top-right corner of the homepage



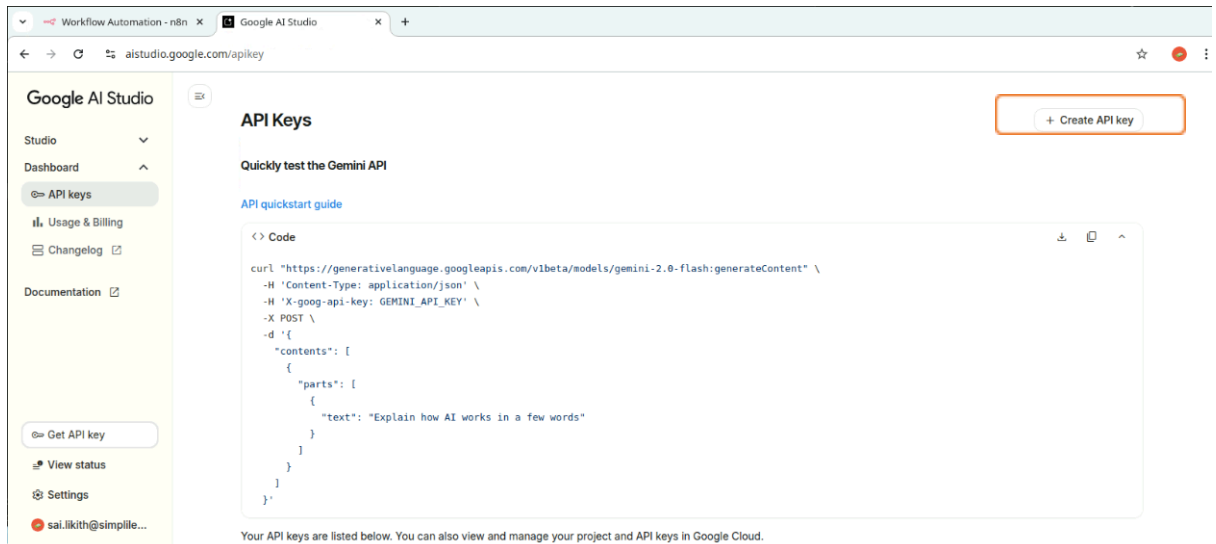
3.3 Sign in using your Google account credentials



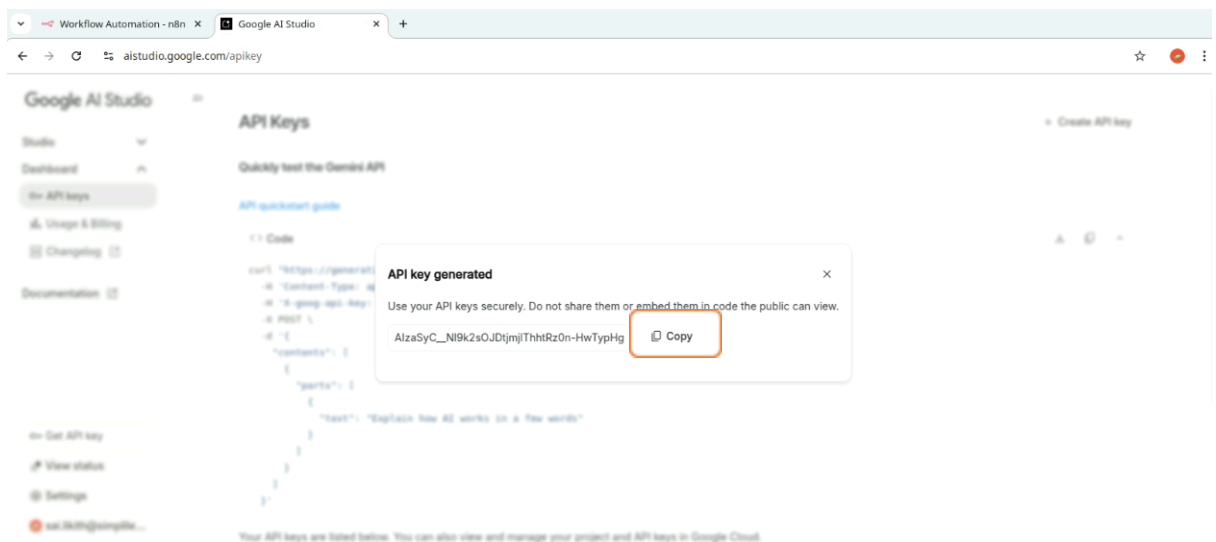
3.4 Once signed in, the dashboard will appear; click the **Get API Key** option at the bottom left.



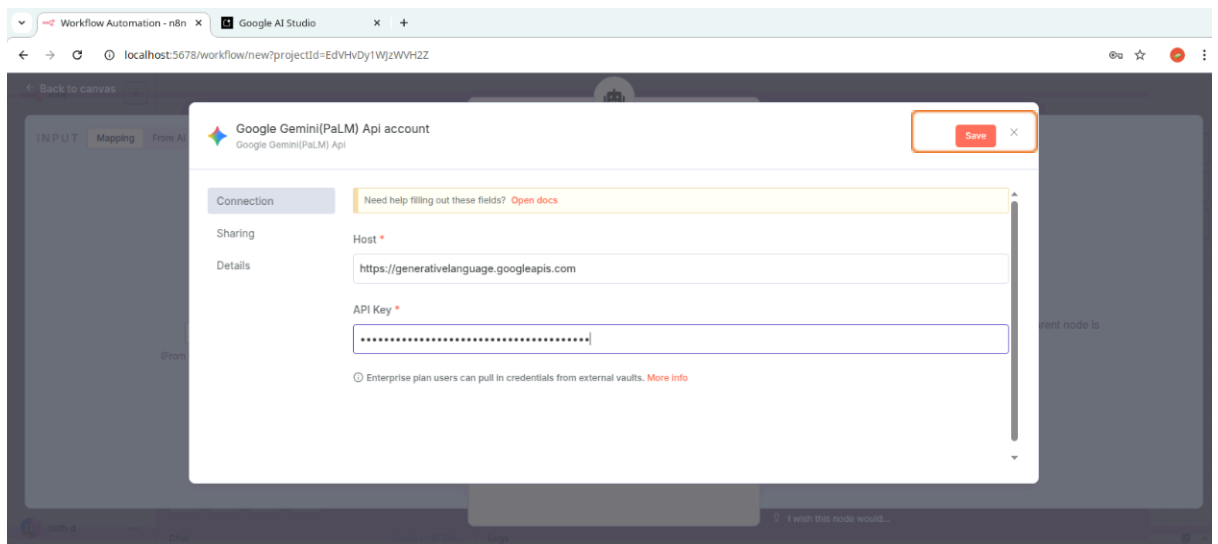
3.5 Click **Create API key** in the top-right corner of the API Keys page



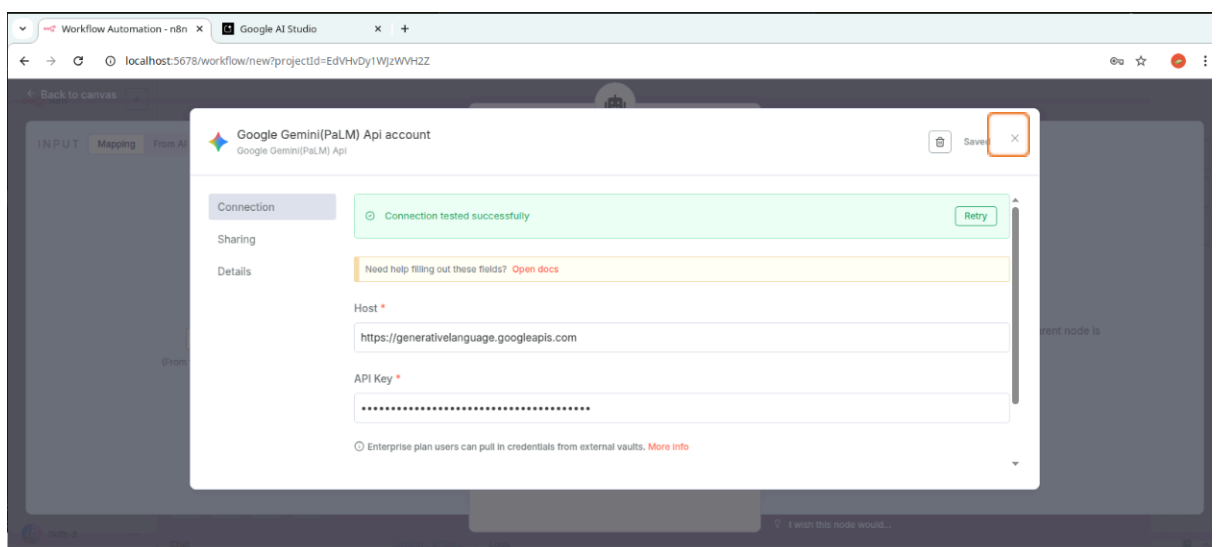
3.6 Copy the API key that is generated



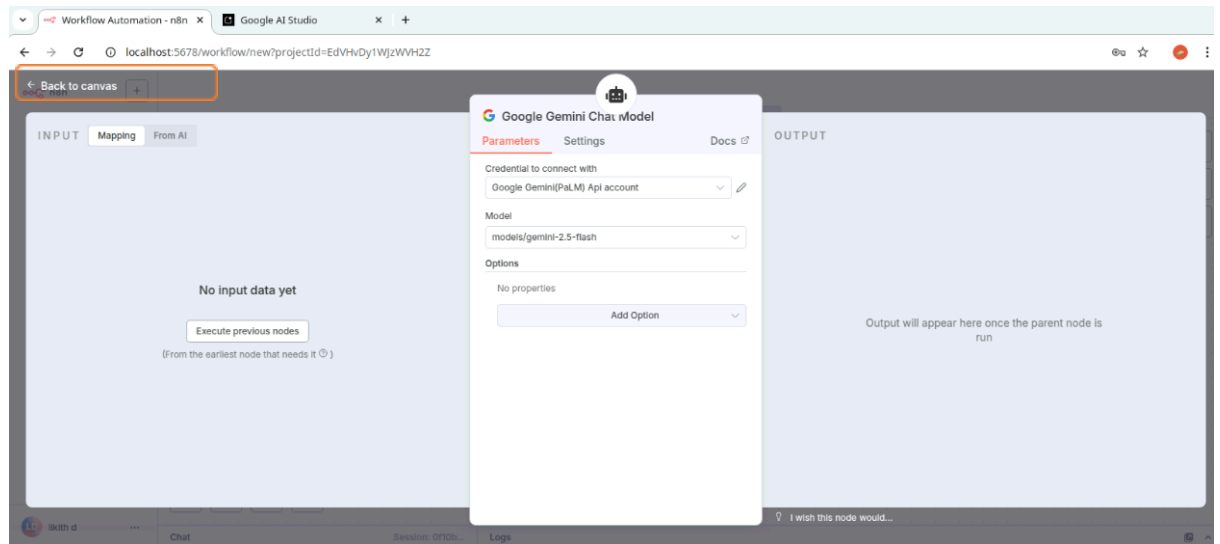
3.7 Return to the n8n interface and paste the API key into the **API Key** field, then click **Save** (top-right corner)



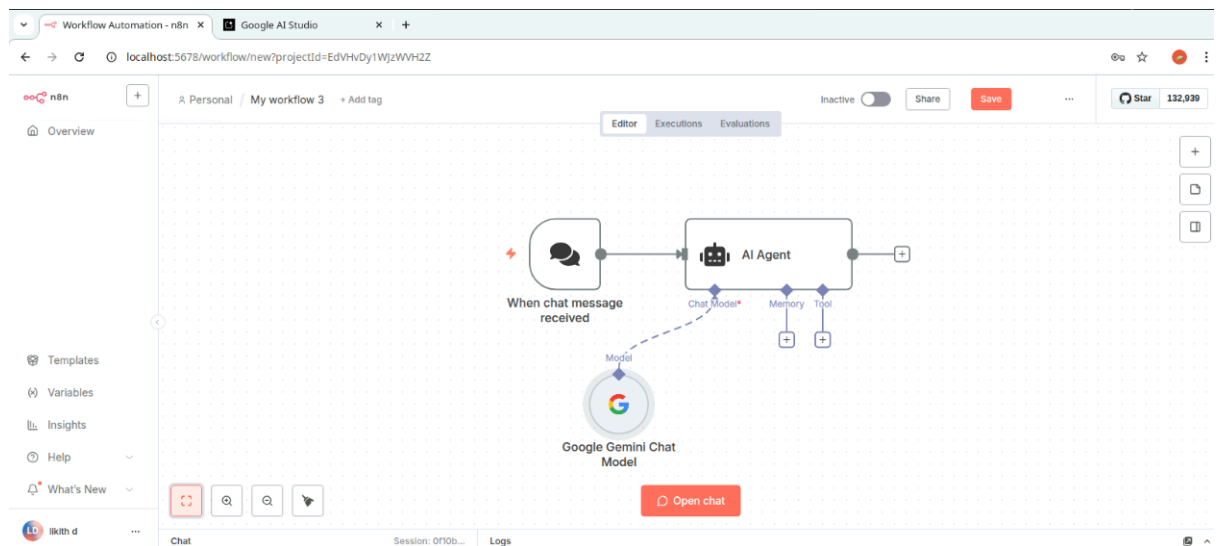
3.8 After saving, close the credentials window by clicking the cross (X) button (top-right corner)



3.9 Finally, click **Back to canvas** in n8n

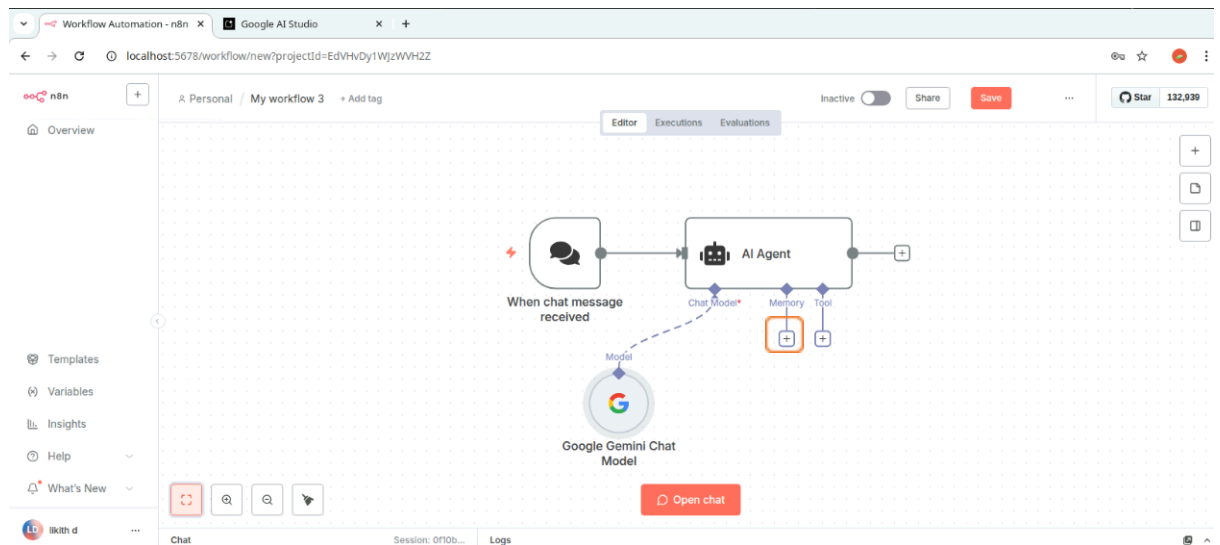


The Google Gemini Chat Model is now successfully integrated with your agent.

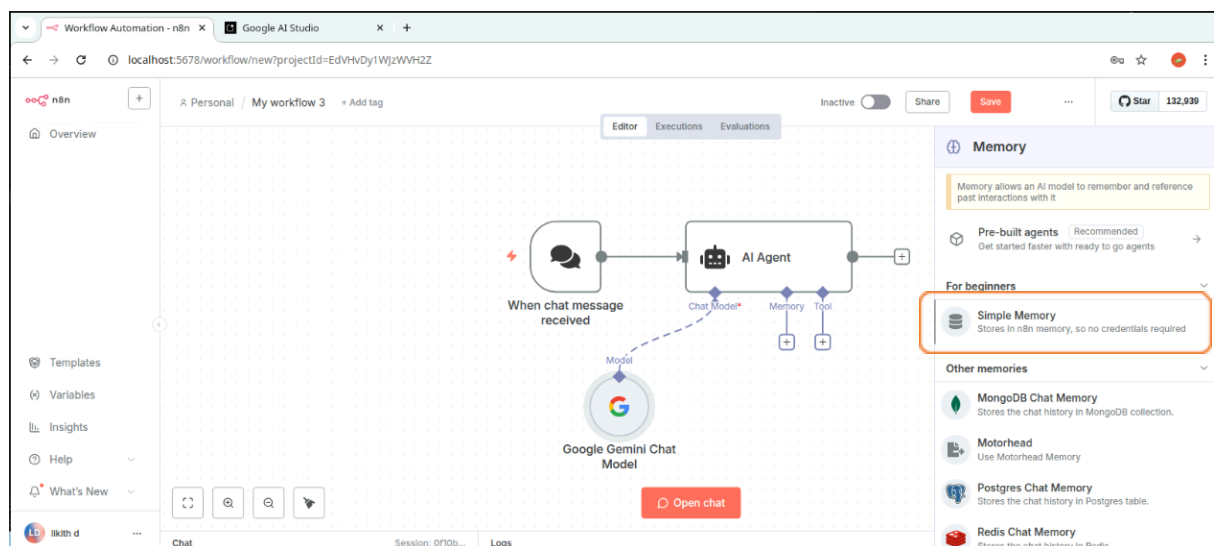


Step 4: Add memory and the MCP Client tool

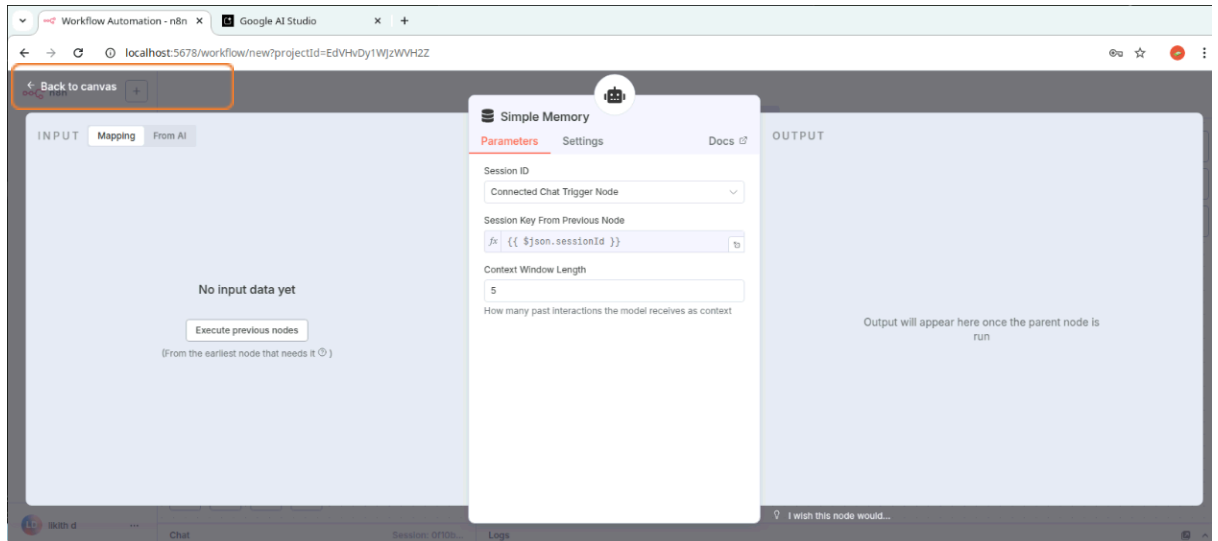
4.1 In the workflow, click the **+** button under the **Memory** section. A menu of memory options will open.



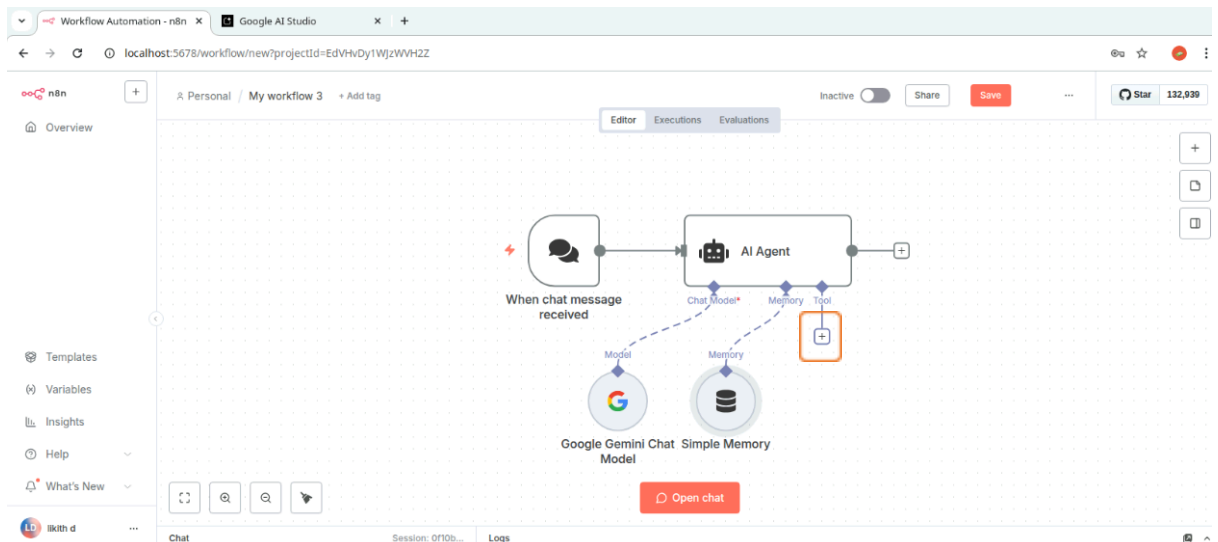
4.2 From the list, select **Simple Memory**



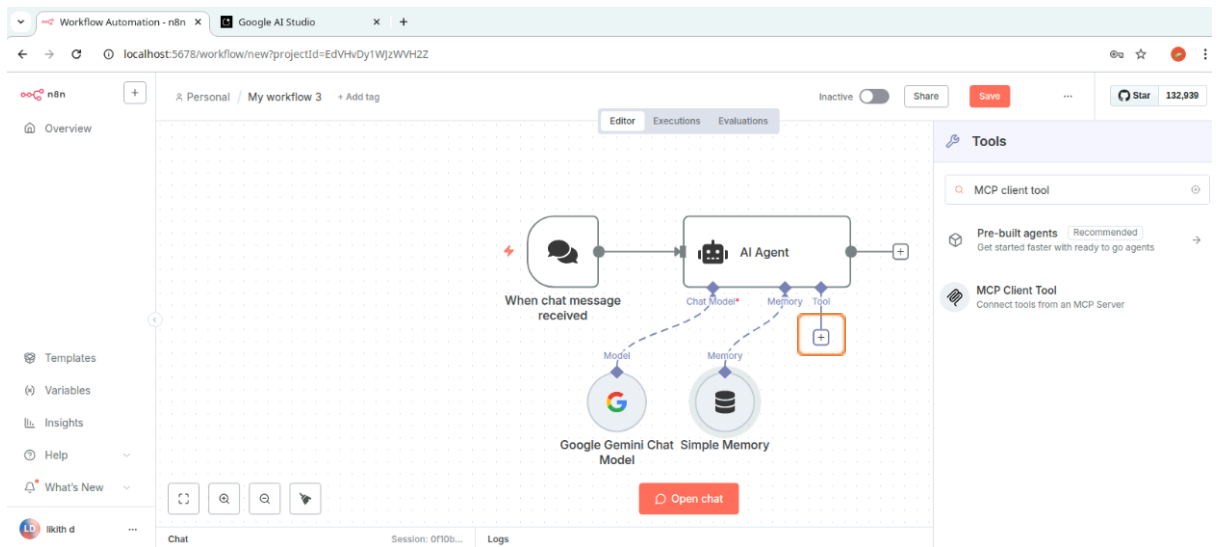
4.3 Once added, click **Back to canvas**



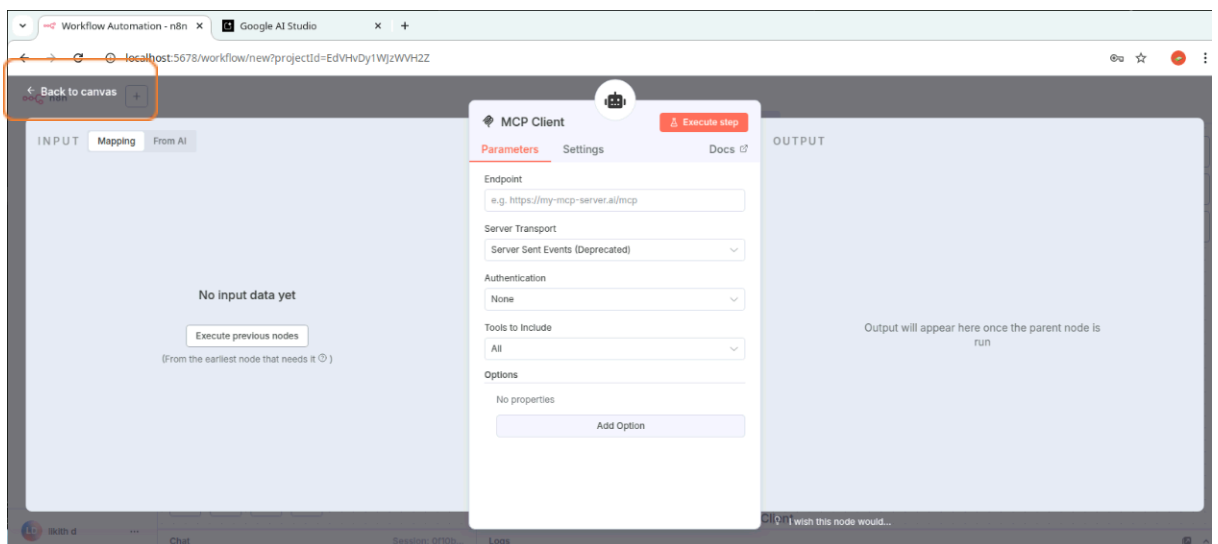
4.4 Next, click the **+** button under the **Tools** section



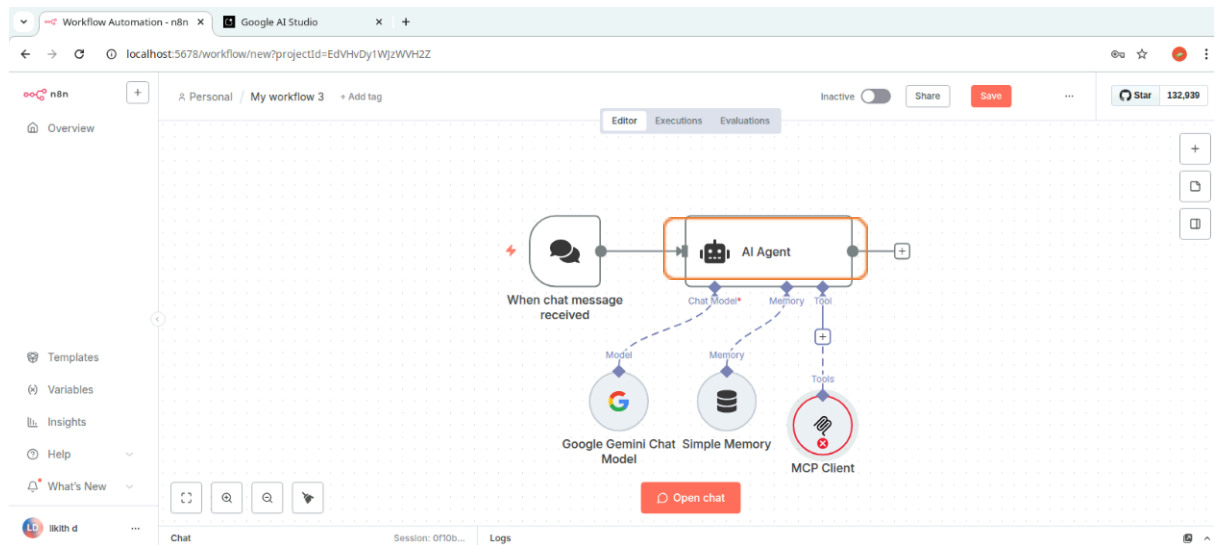
4.5 In the search bar, type **MCP Client Tool** and select it



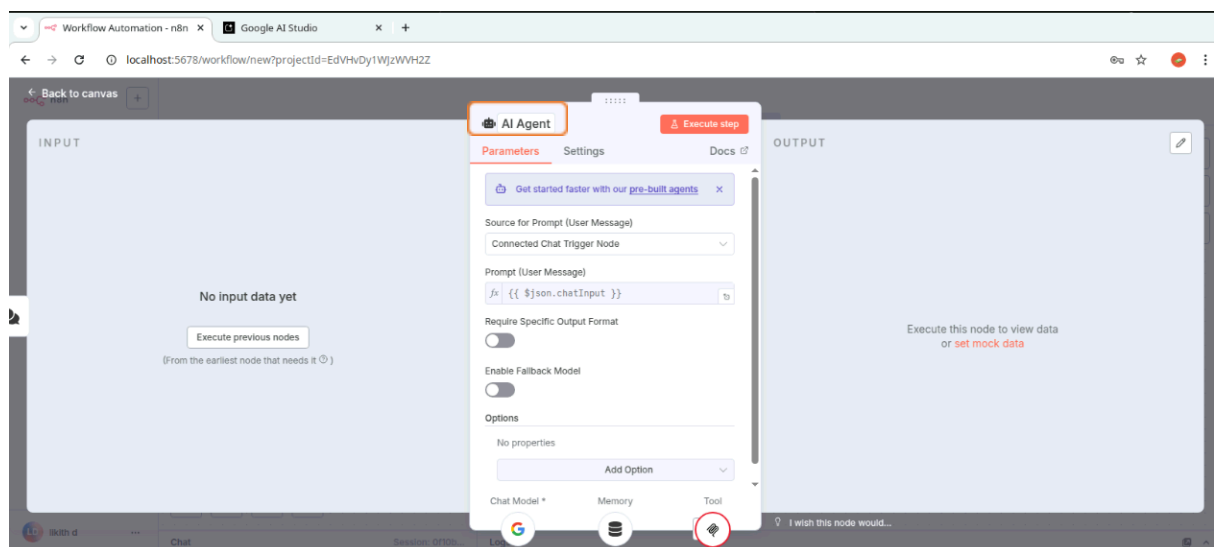
4.6 After it is added, click **Back to canvas**

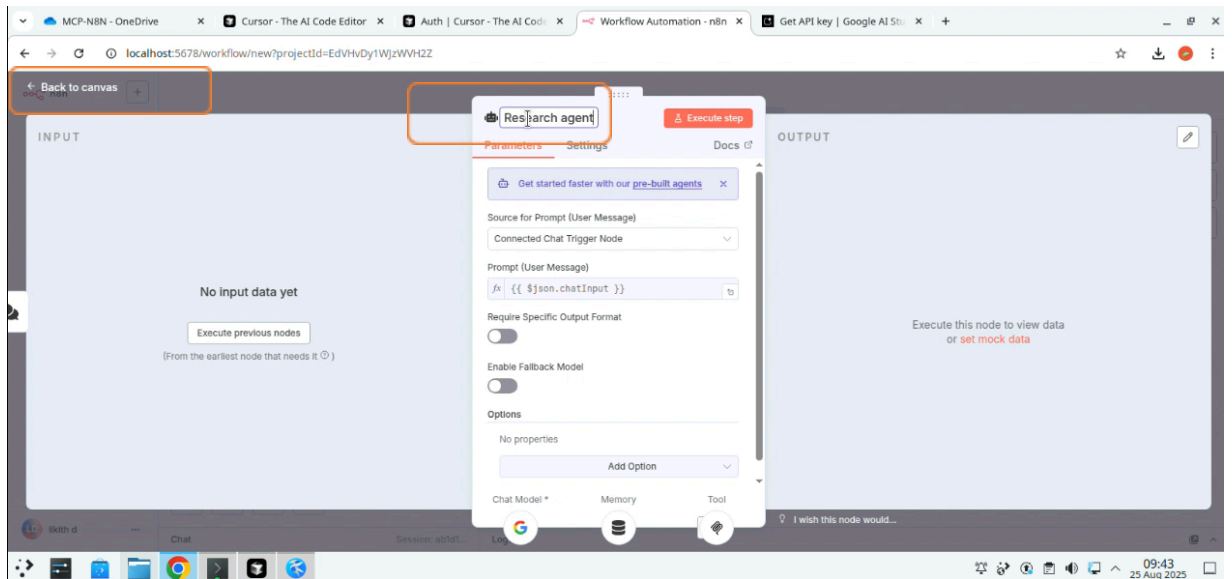


4.7 Double-click the **AI Agent** node to open its settings

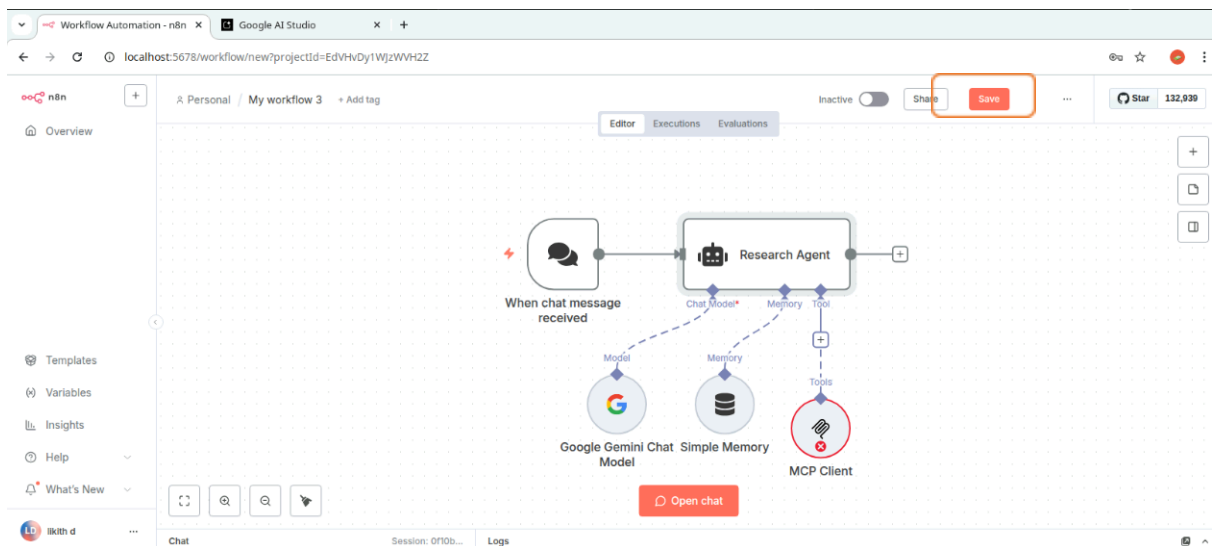


4.8 Change the name of the agent by double-clicking on the AI Agent option (top left) and replacing the default name with Research Agent. Then, click on **Back to canvas**.



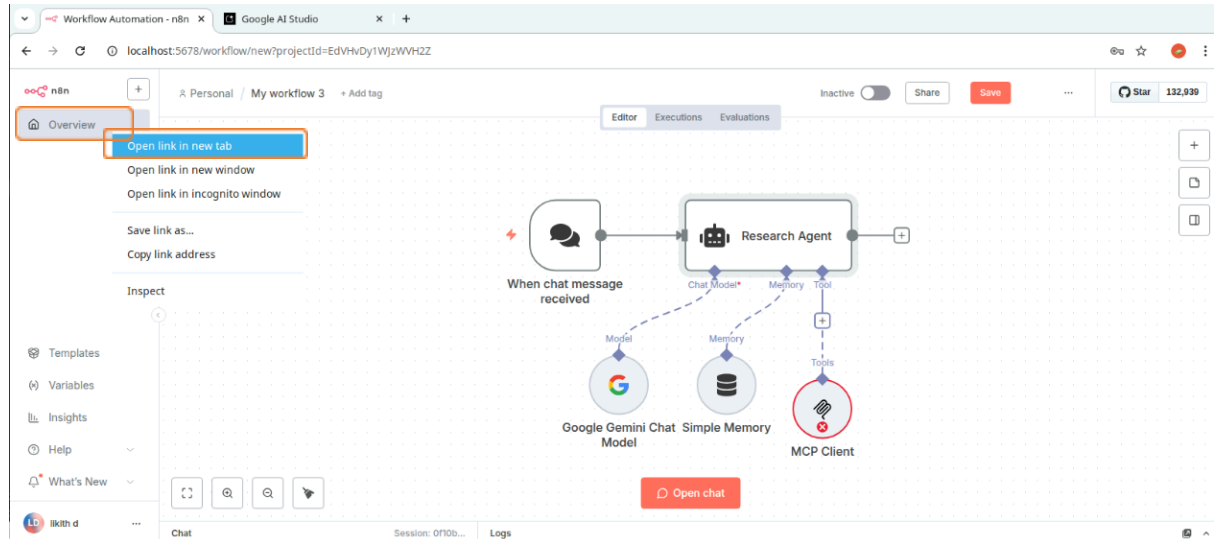


4.9 Click the **Save** button (top right)

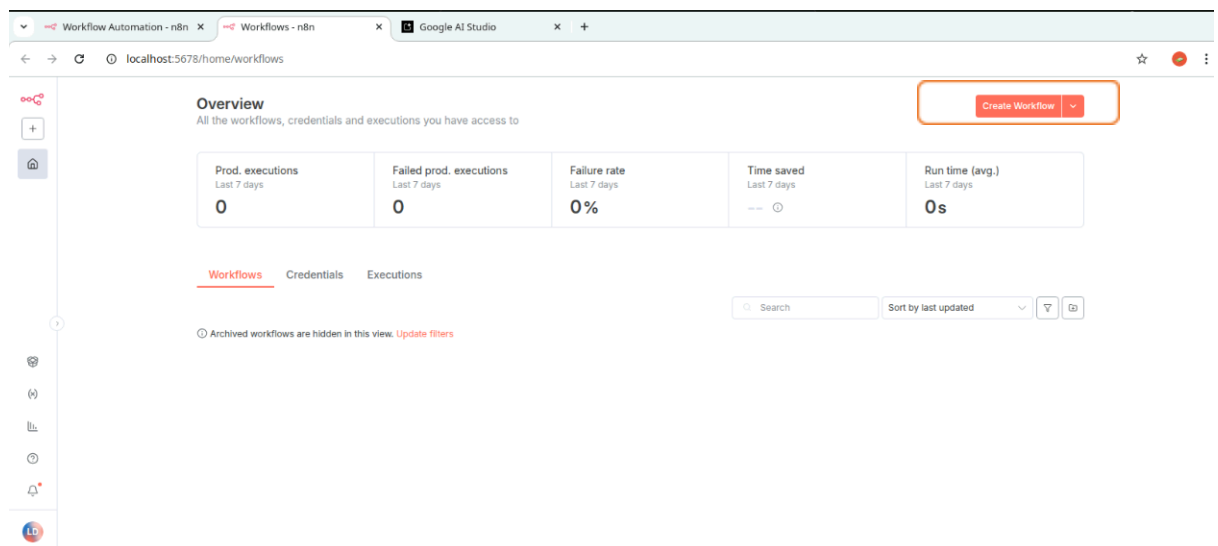


Step 5: Create an MCP server trigger in n8n

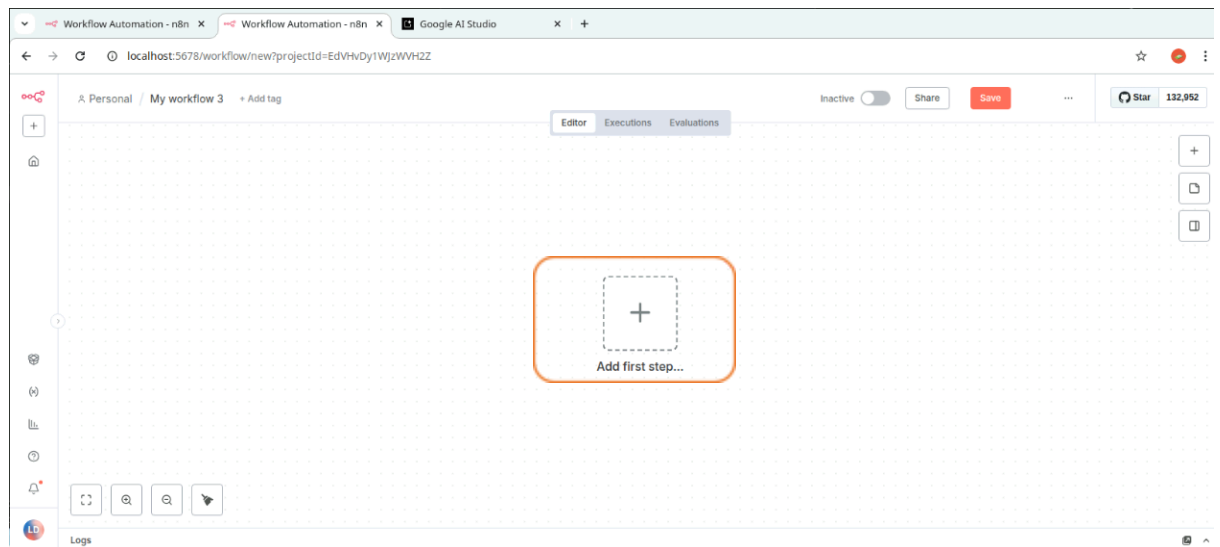
5.1 In the n8n dashboard, right-click on the **Overview** option and select **Open link in new tab**



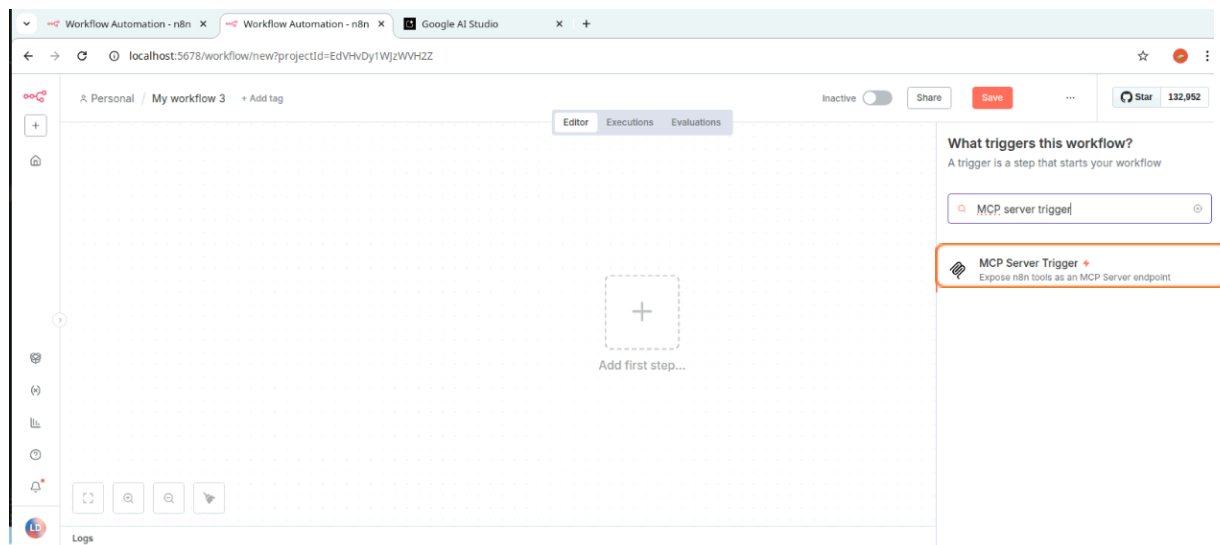
5.2 In the newly opened tab, click the **Create Workflow** button



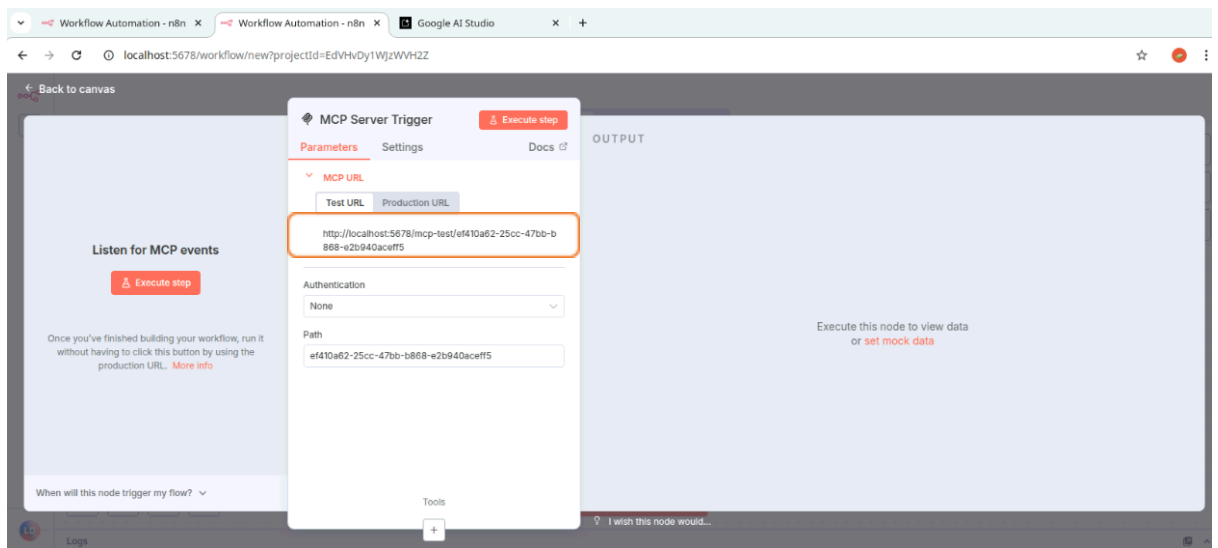
5.3 Inside the workflow editor, click the + button to add a new node. A search bar will appear on the right side.



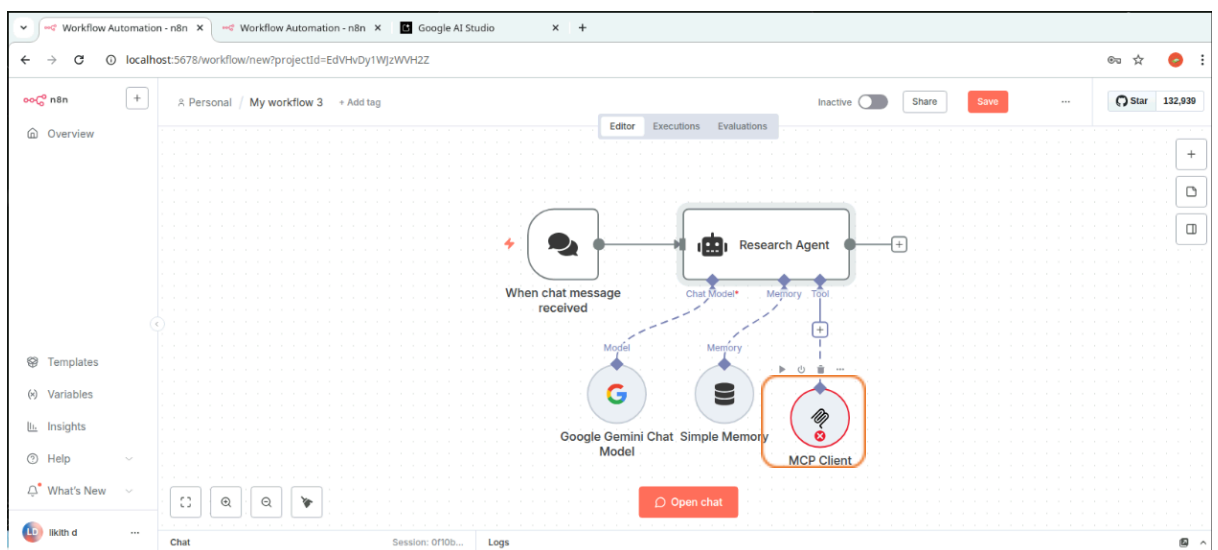
5.4 In the search bar, type **MCP Server Trigger** and select it from the list



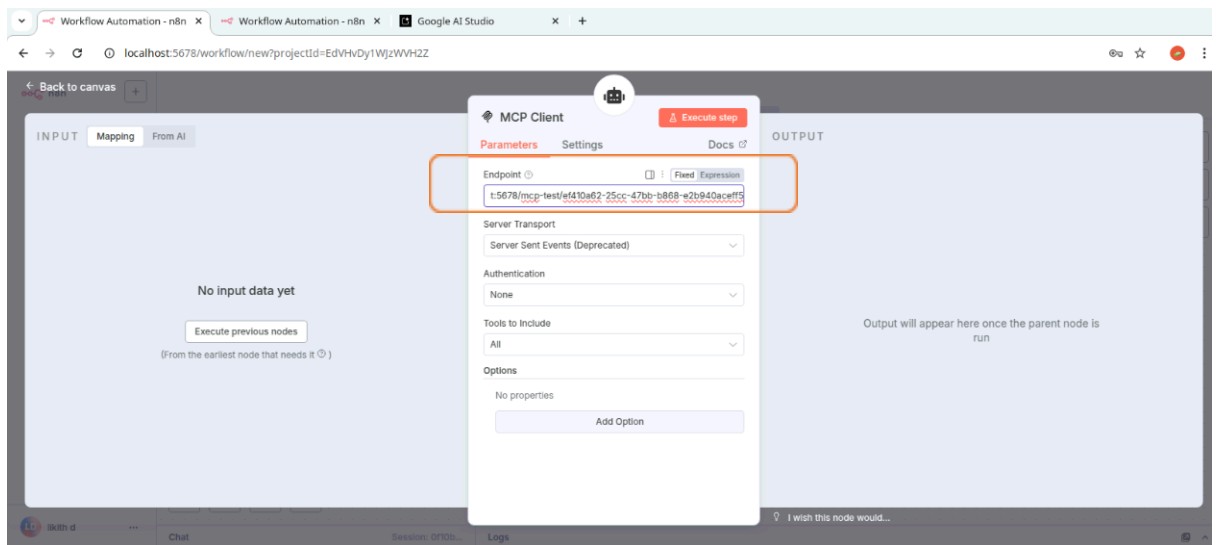
5.5 Copy the URL provided in the MCP Server Trigger node



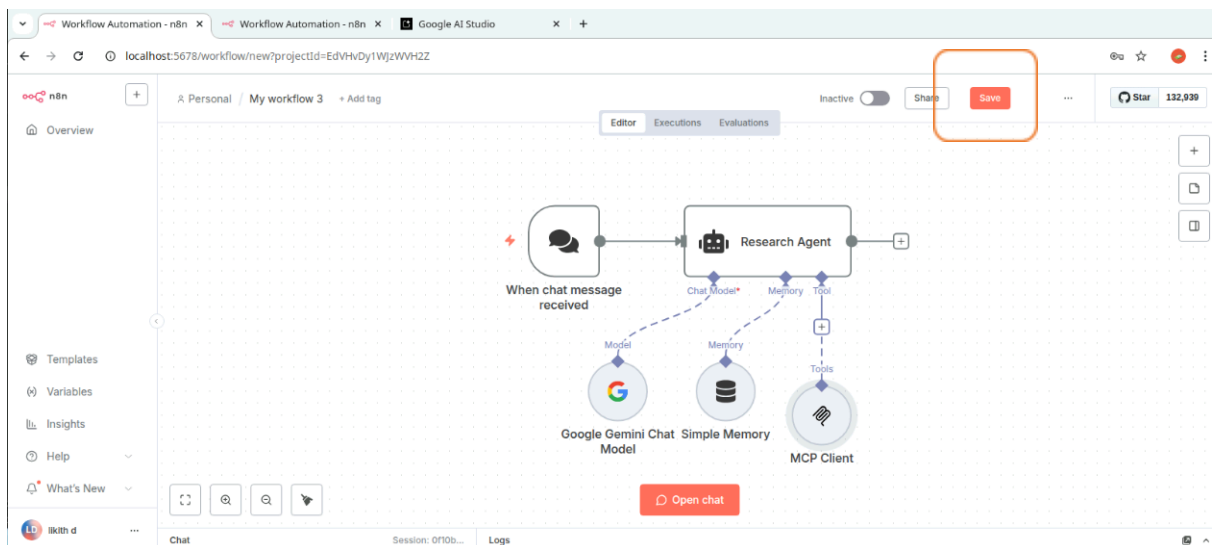
5.6 Switch back to the agent workflow tab and double-click the MCP Client trigger node



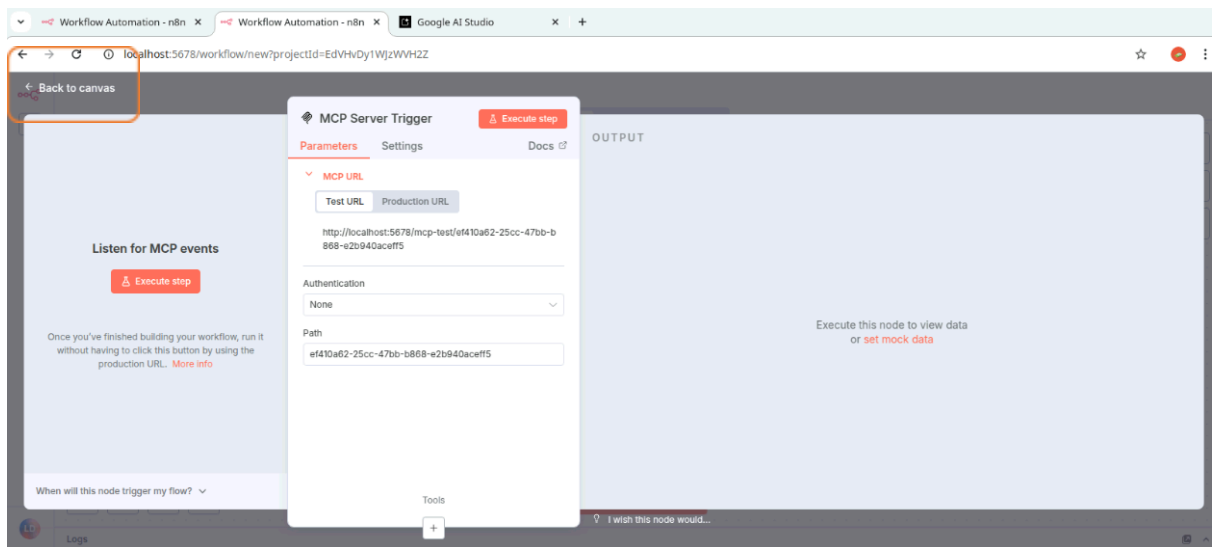
5.7 Paste the copied URL into the **Endpoint** field, then click **Back to canvas**



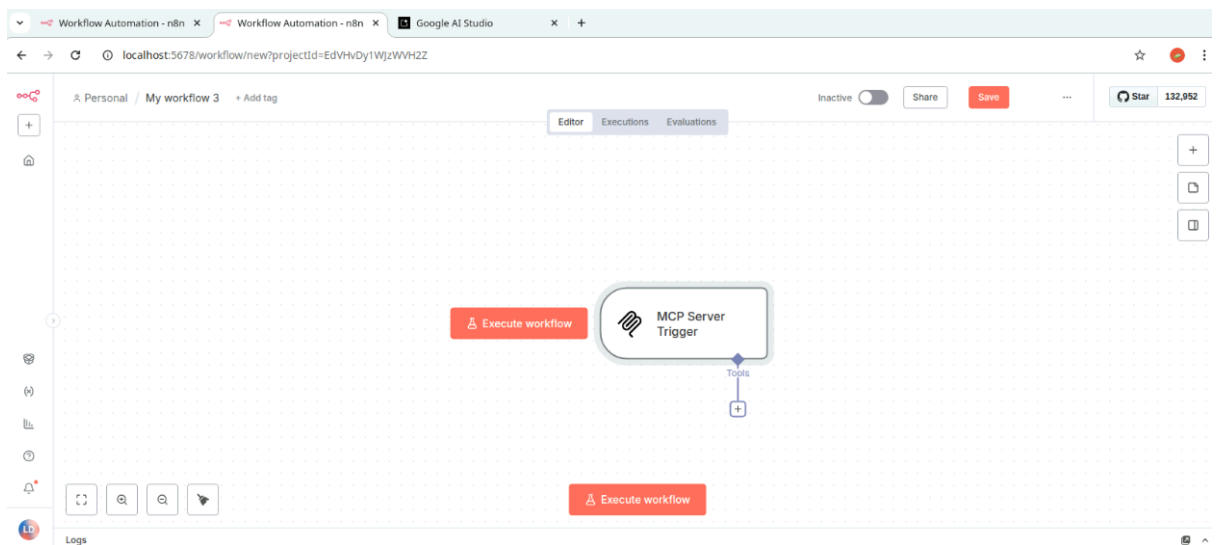
5.8 Save the workflow by clicking the **Save** button (top-right corner)



5.9 Return to the MCP Server Workflow tab and click **Back to canvas**

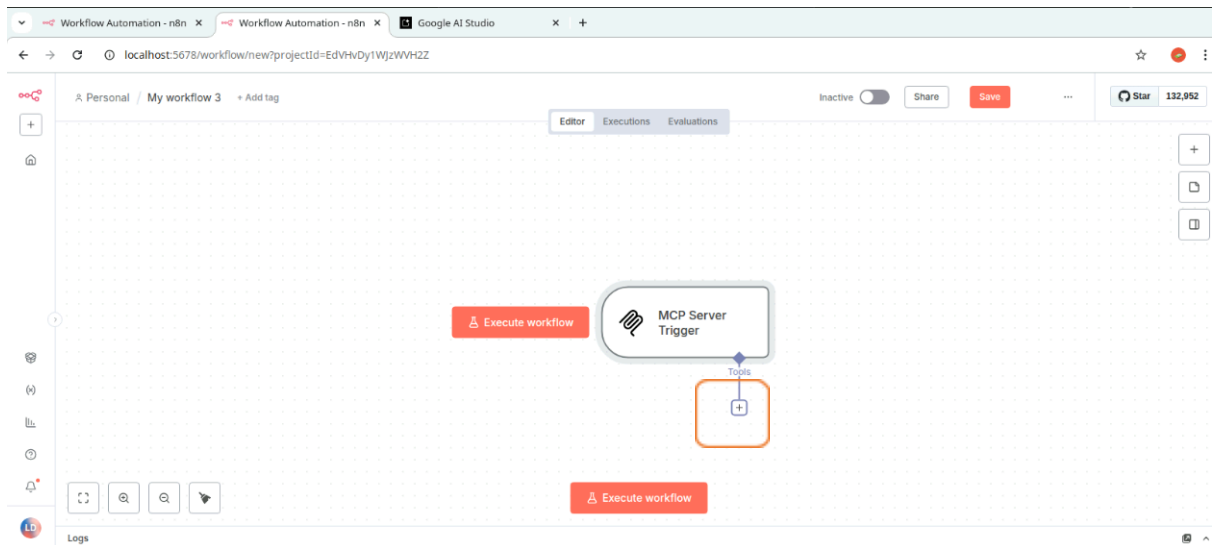


Output:

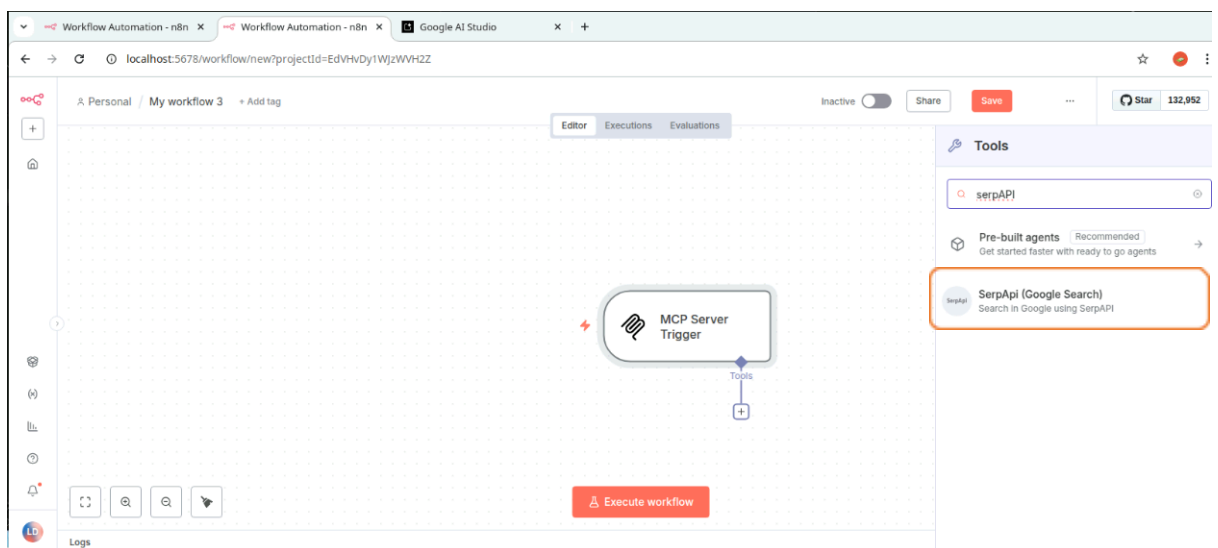


Step 6: Integrate the SerpApi tool

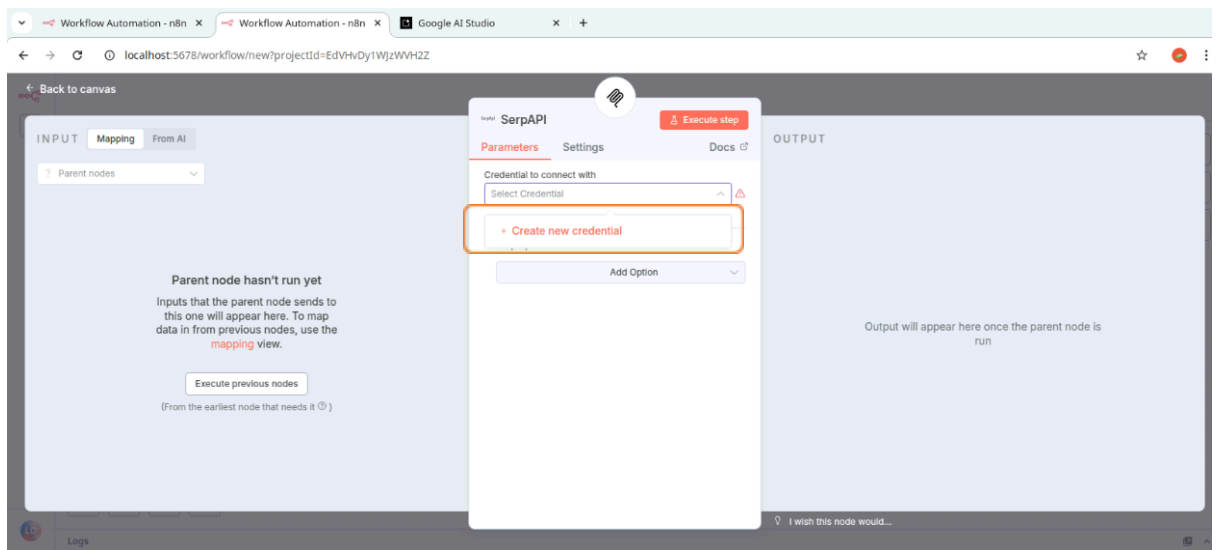
6.1 In the MCP Server Trigger workflow, click the + button under the **Tools** section



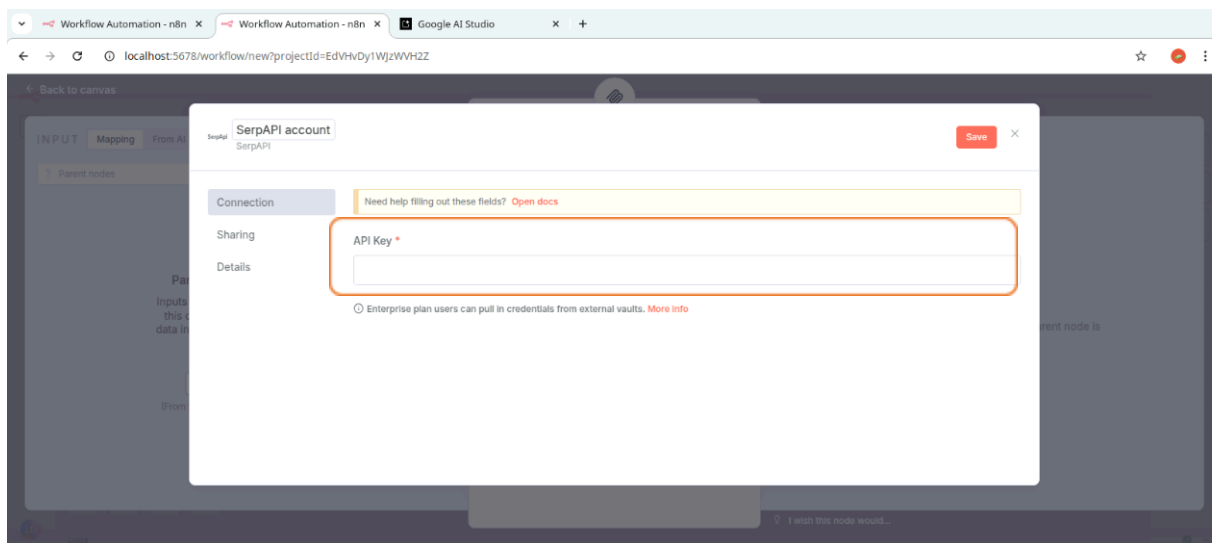
6.2 In the search bar, type **SerpApi** and select **SerpApi (Google Search)**



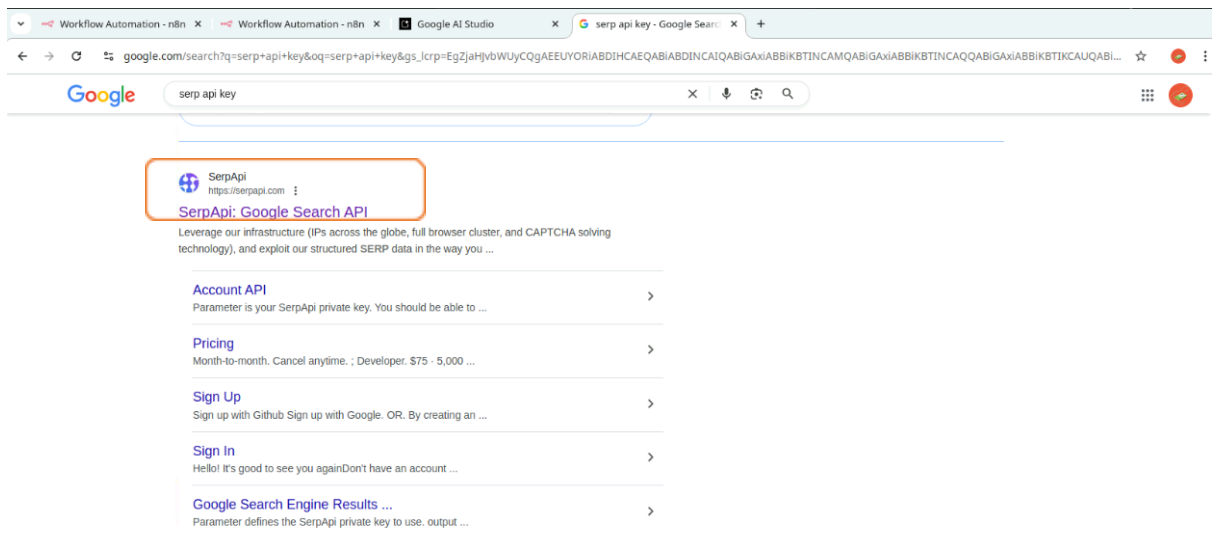
6.3 In the credentials section, click **Create new credential**



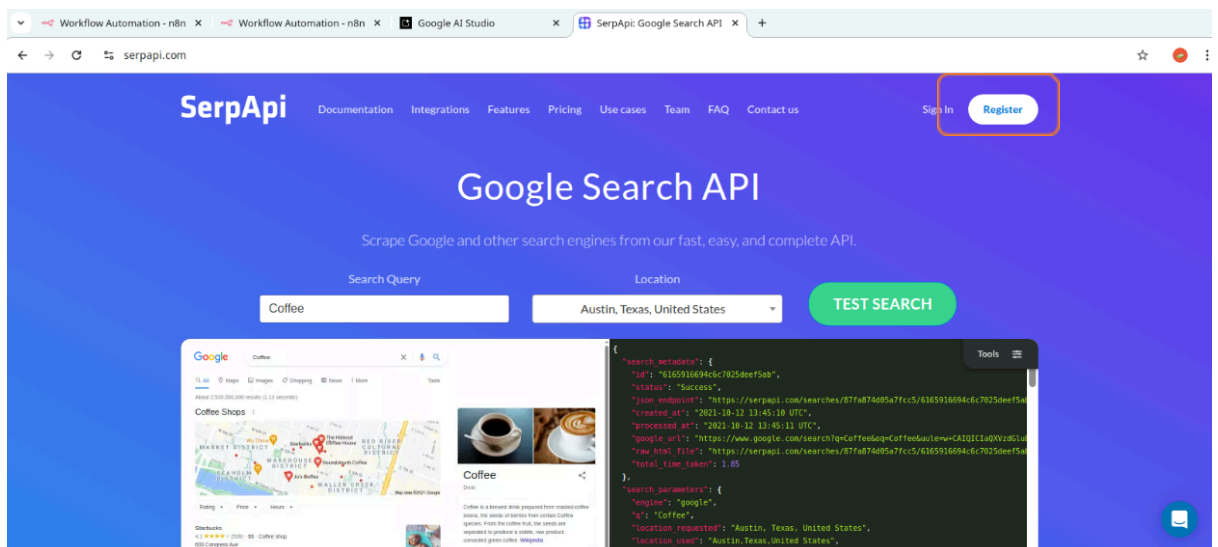
This action opens the API key window.



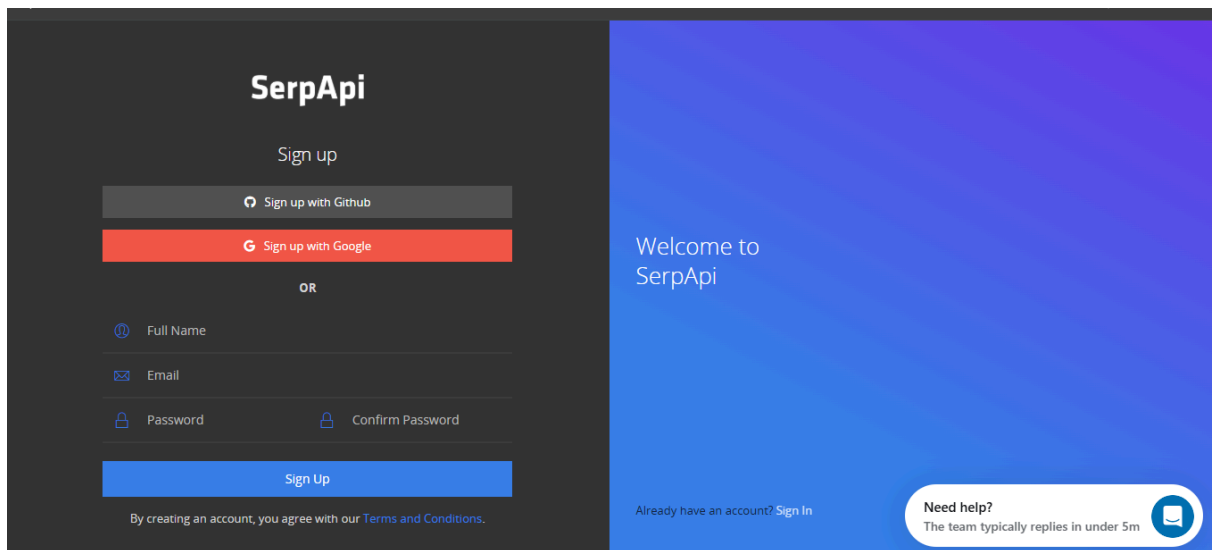
6.4 Open a new browser window and search for **SerpApi Key**. From the results, open the SerpApi website.



6.5 Click on the website and then select register



6.6 Sign up using your email



The image shows the SerpApi sign-up page. On the left, a dark grey sidebar contains the SerpApi logo and a 'Sign up' heading. Below this are two buttons: 'Sign up with Github' and 'Sign up with Google'. A red 'OR' separator follows. Then, there are input fields for 'Full Name', 'Email', 'Password', and 'Confirm Password'. A blue 'Sign Up' button is at the bottom of the form. A small text line reads: 'By creating an account, you agree with our [Terms and Conditions](#).' On the right, a blue gradient background features the text 'Welcome to SerpApi'. At the bottom right, there is a 'Need help?' section with the text 'The team typically replies in under 5m' and a chat icon. A link 'Already have an account? Sign In' is also present.

SerpApi

Sign up

[Sign up with Github](#)

[Sign up with Google](#)

OR

Full Name

Email

Password Confirm Password

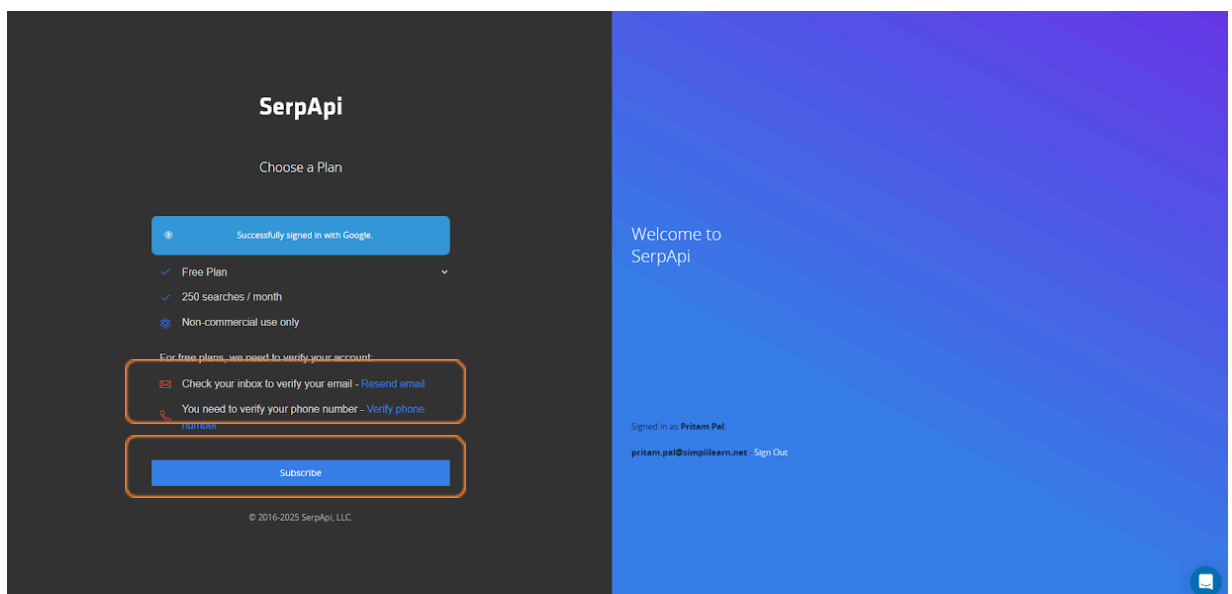
[Sign Up](#)

By creating an account, you agree with our [Terms and Conditions](#).

Already have an account? [Sign In](#)

Need help?
The team typically replies in under 5m

6.7 After signing up, complete the verification process for both your email and mobile number. Once verified, click **Subscribe**.



The image shows the SerpApi 'Choose a Plan' page. The left sidebar has the SerpApi logo and 'Choose a Plan' heading. A blue box says 'Successfully signed in with Google'. Below are plan details: 'Free Plan', '250 searches / month', and 'Non-commercial use only'. A red box highlights verification steps: 'Check your inbox to verify your email - [Resend email](#)' and 'You need to verify your phone number - [Verify phone number](#)'. A blue 'Subscribe' button is at the bottom. The right side has a blue gradient background with 'Welcome to SerpApi'. At the bottom, it says 'Signed in as Pritam Pal.' and 'pritam.pal@simplelearn.net Sign Out'. A chat icon is in the bottom right corner.

SerpApi

Choose a Plan

Successfully signed in with Google

✓ Free Plan

✓ 250 searches / month

⚙ Non-commercial use only

For free plans, we need to verify your account.

✉ Check your inbox to verify your email - [Resend email](#)

📞 You need to verify your phone number - [Verify phone number](#)

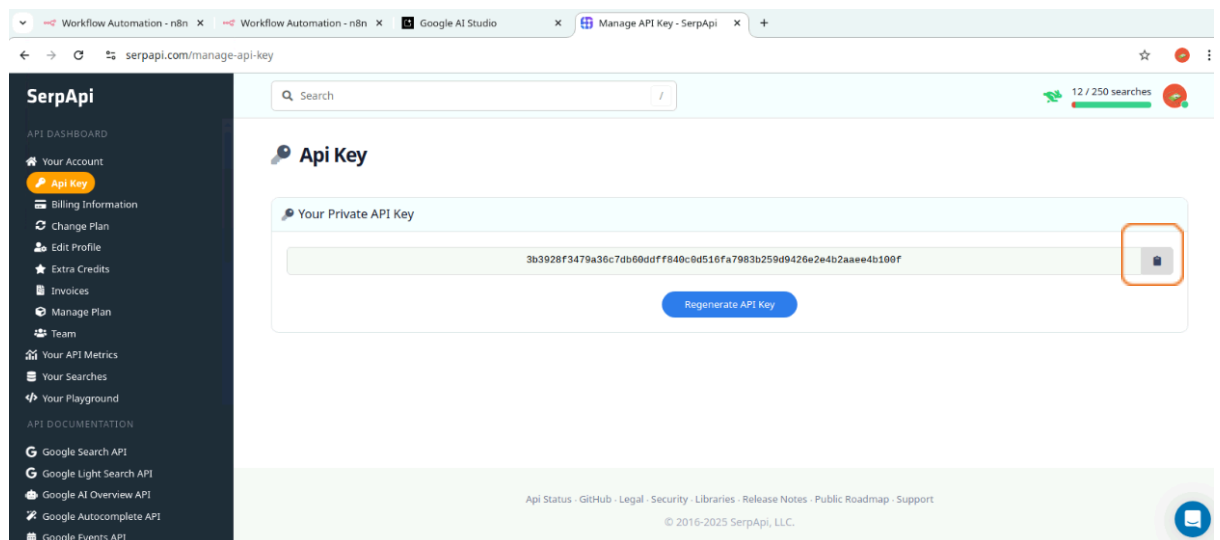
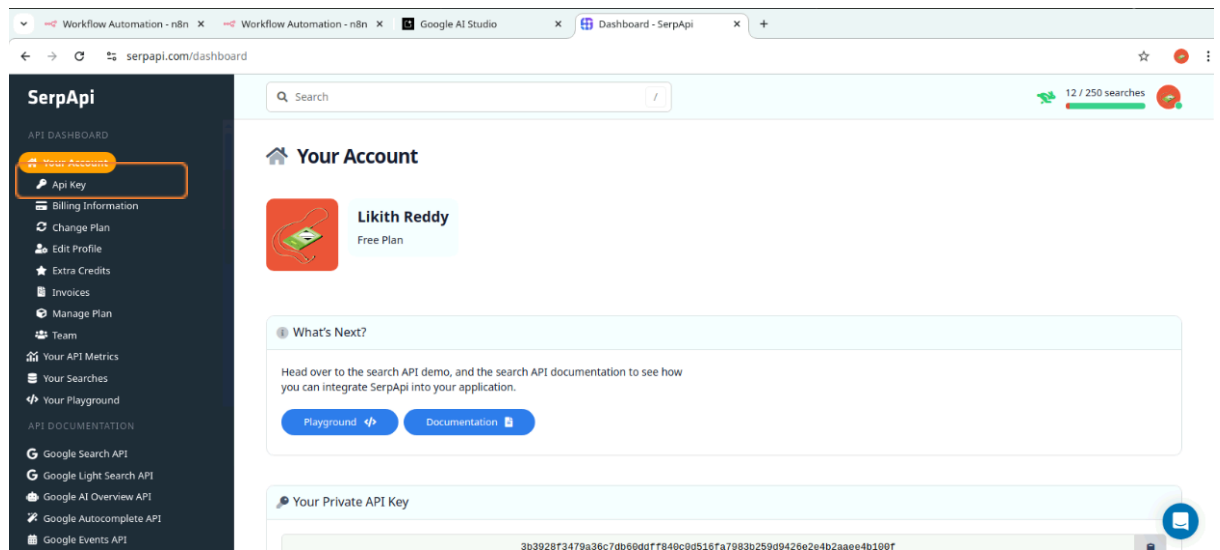
[Subscribe](#)

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Welcome to SerpApi

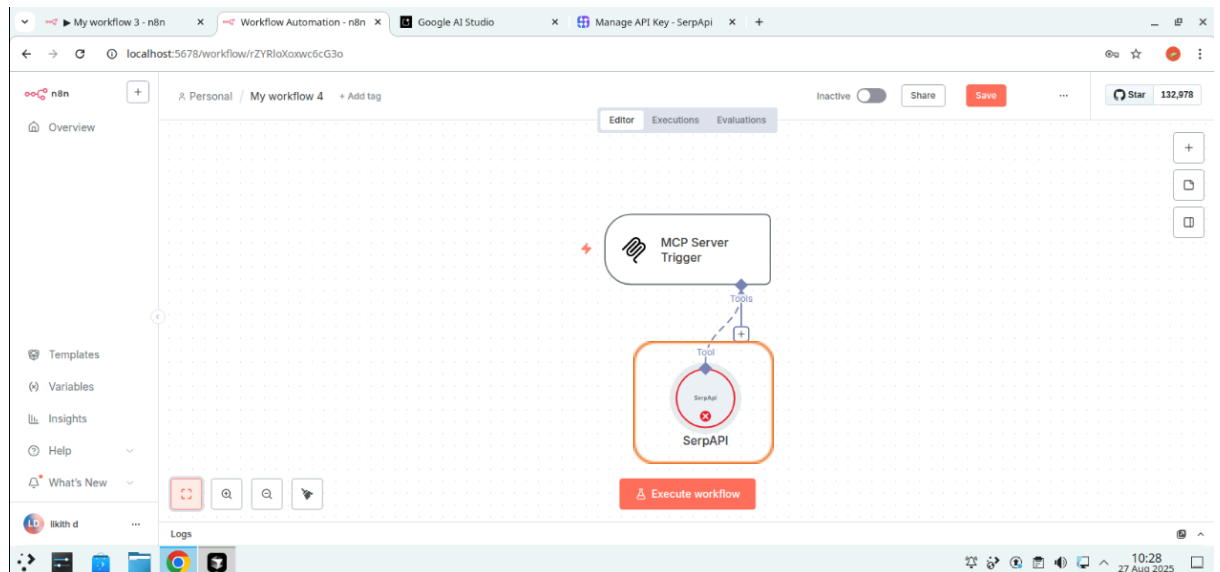
Signed in as Pritam Pal.
[pritam.pal@simplelearn.net](#) [Sign Out](#)

6.8 On the SerpApi dashboard, click **API Key** (left-hand side) and copy the generated key

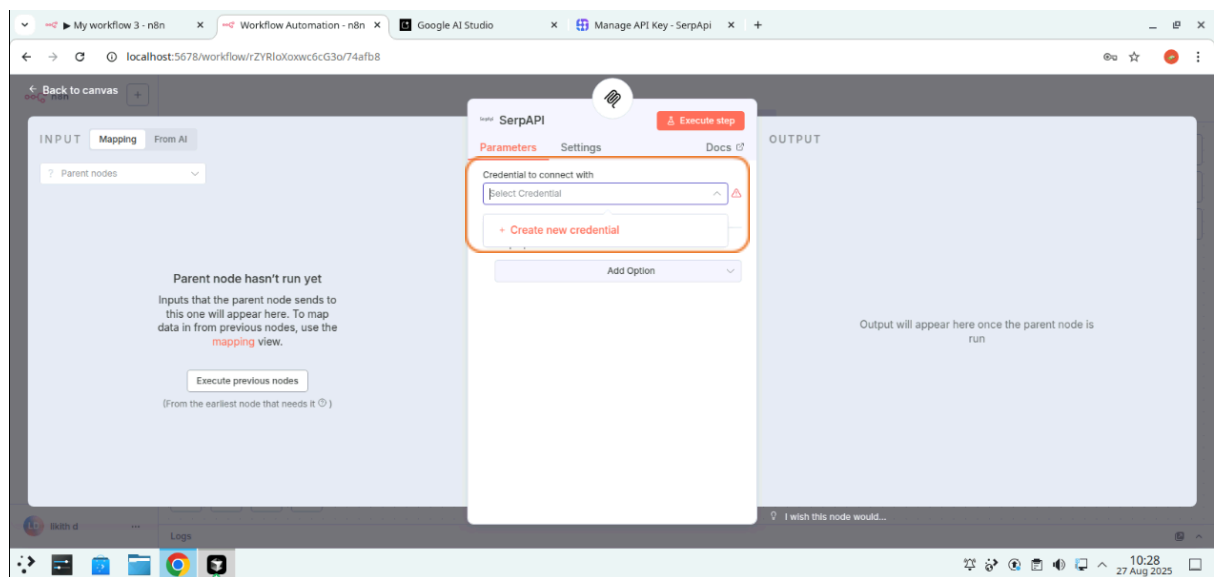


Step 7: Add the SerpApi key and activate the server trigger

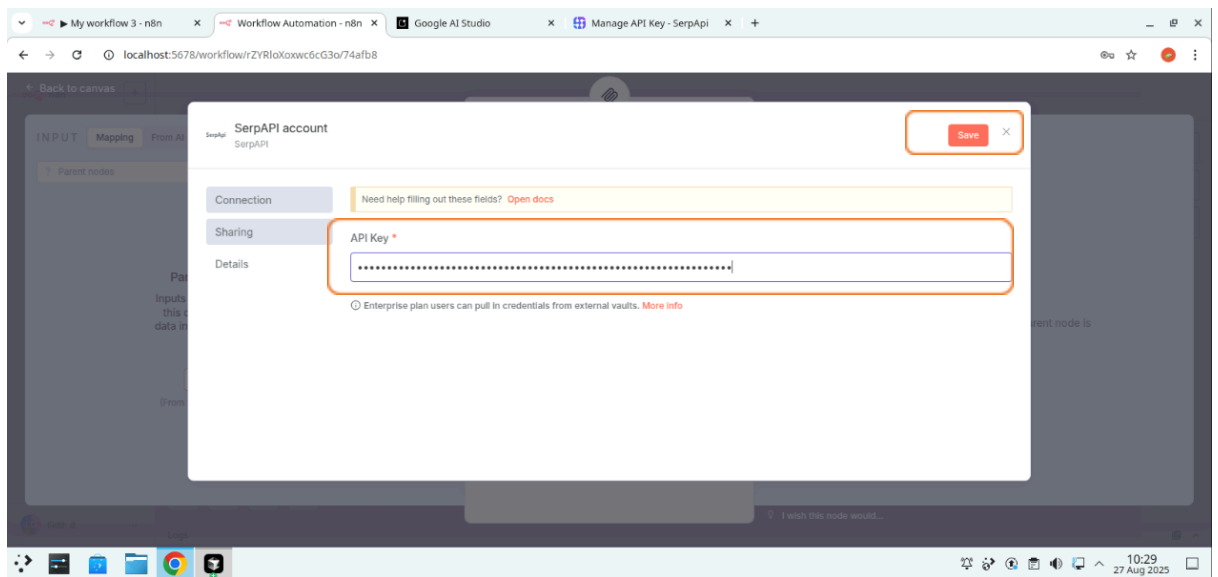
7.1 Go back to the **MCP Server Trigger** workflow and double-click the **SerpAPI** node



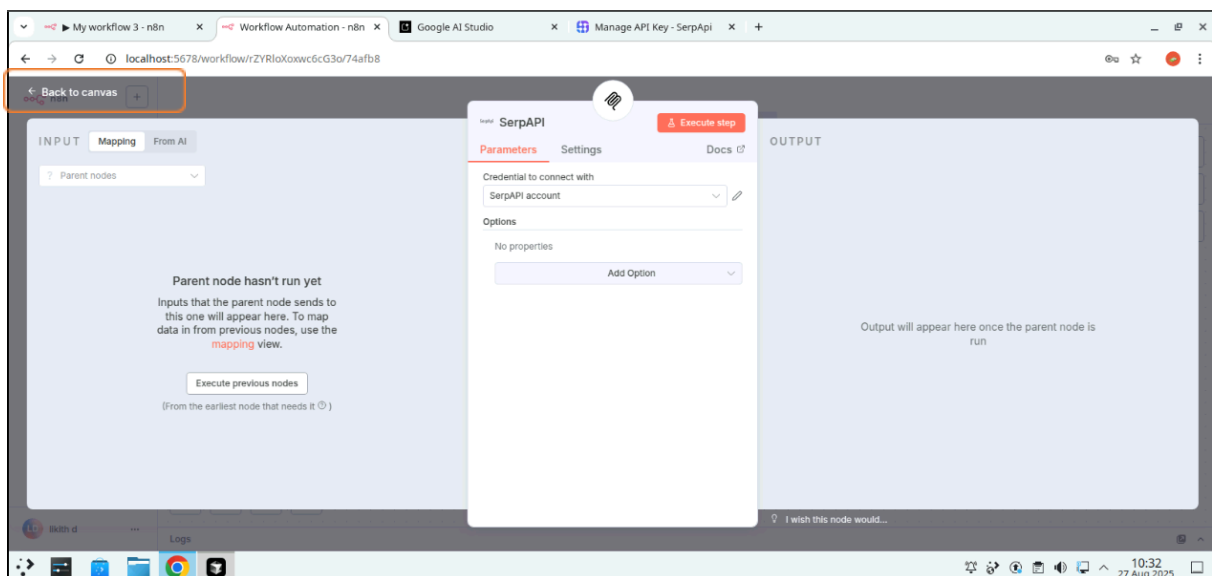
7.2 In the credentials section, click **Create new credential**



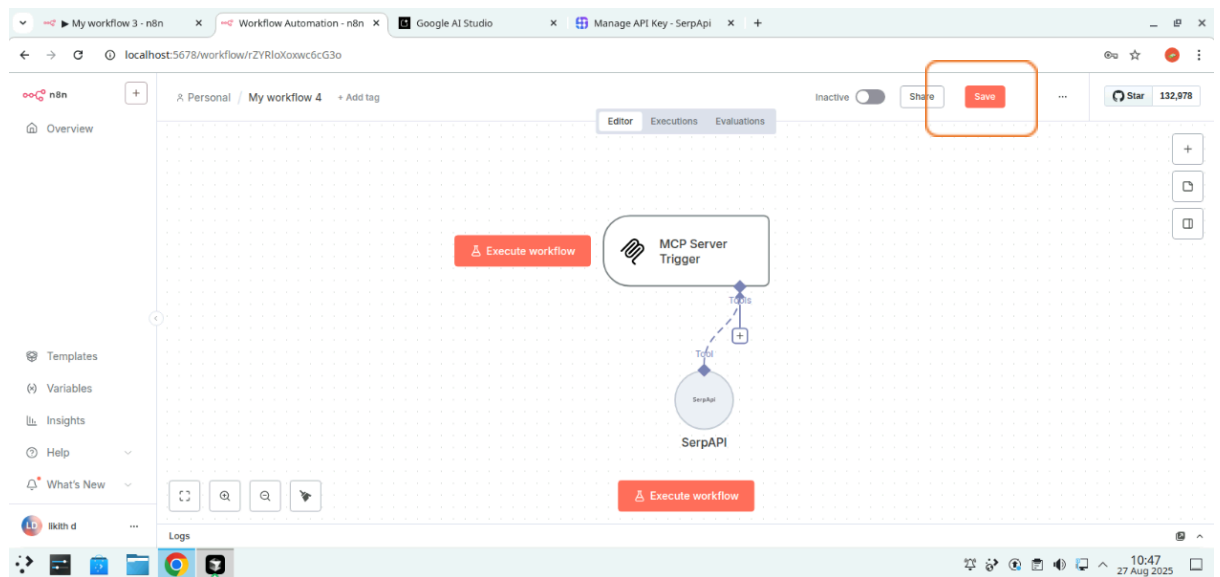
7.3 Paste the copied SerpApi key into the **API Key** field and click **Save** at the top right. Close the credentials window by clicking the cross (X) button (top-right corner).



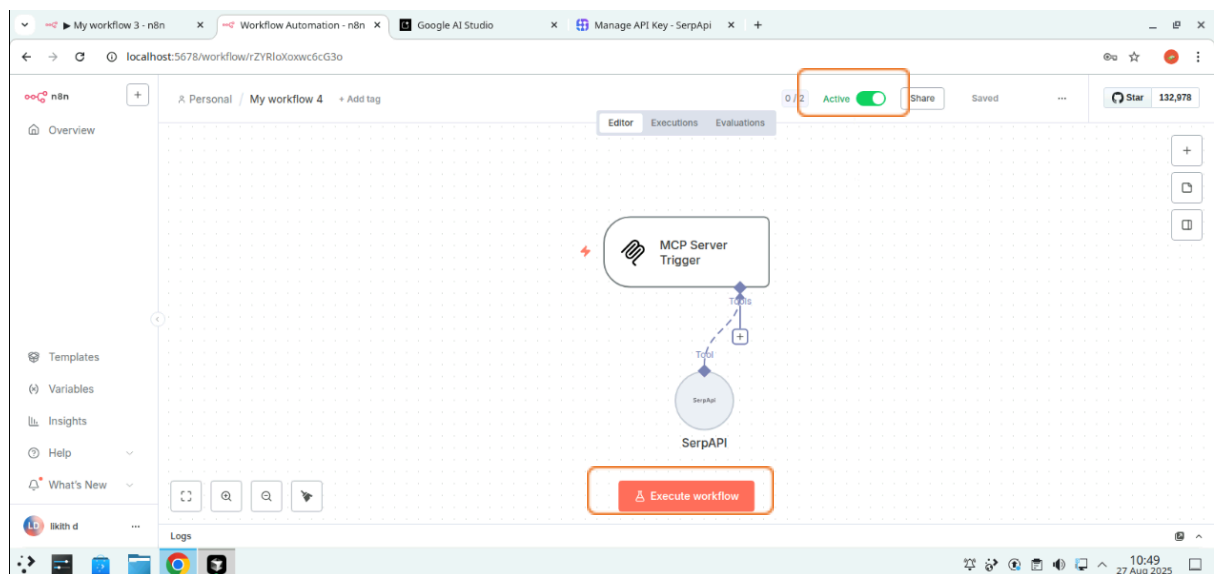
7.4 Click **Back to canvas** (top left)



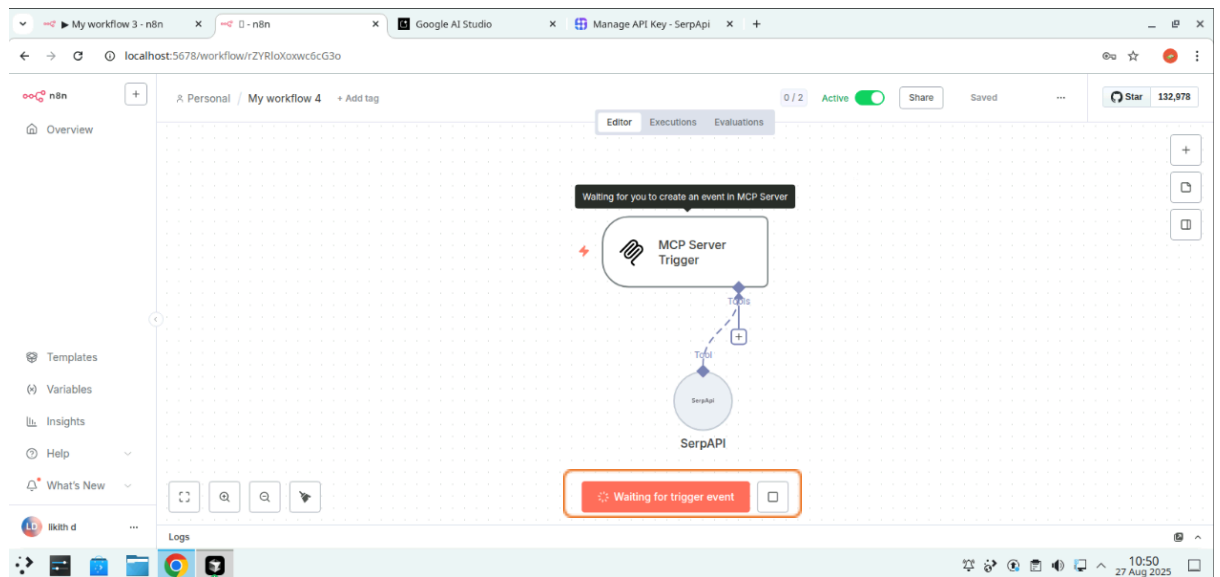
7.5 Save the workflow by clicking the **Save** button (top right)



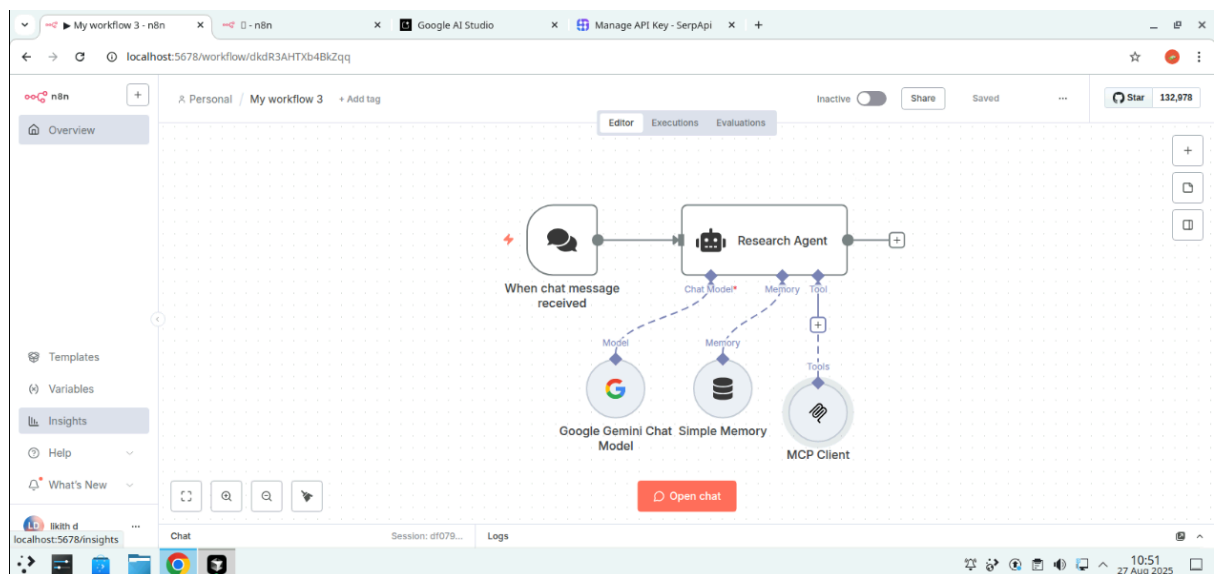
7.6 Activate the server by switching the status from **Inactive** to **Active**, then click the **Execute Workflow** button at the bottom



The workflow will now show the status as **Waiting for trigger event**, indicating it is ready to receive input.

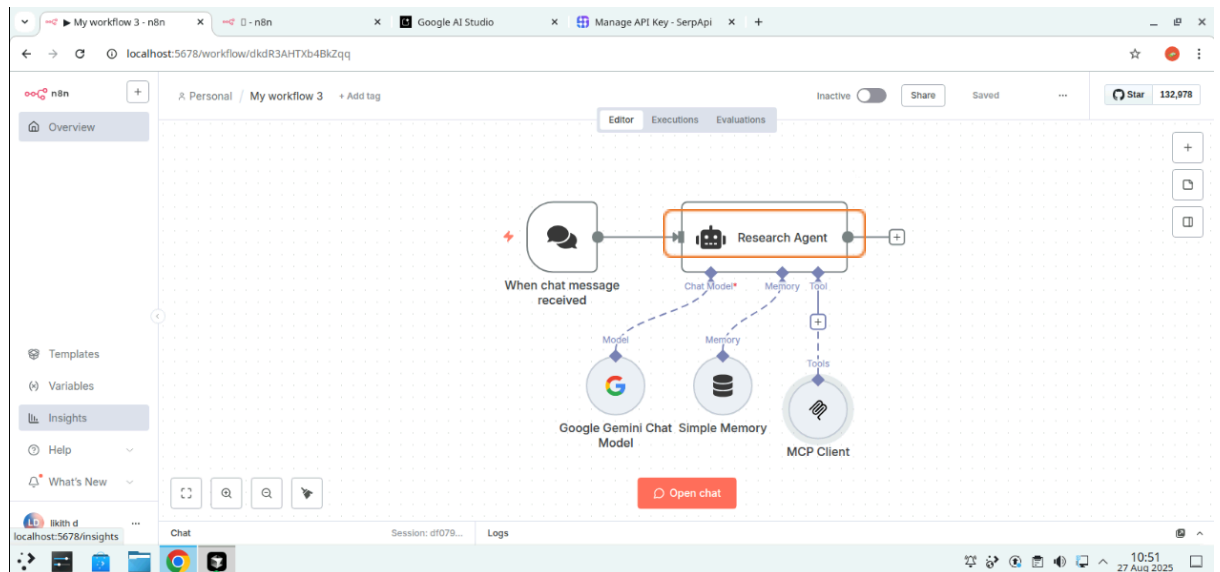


7.7 Now get back to the Research Agent workflow

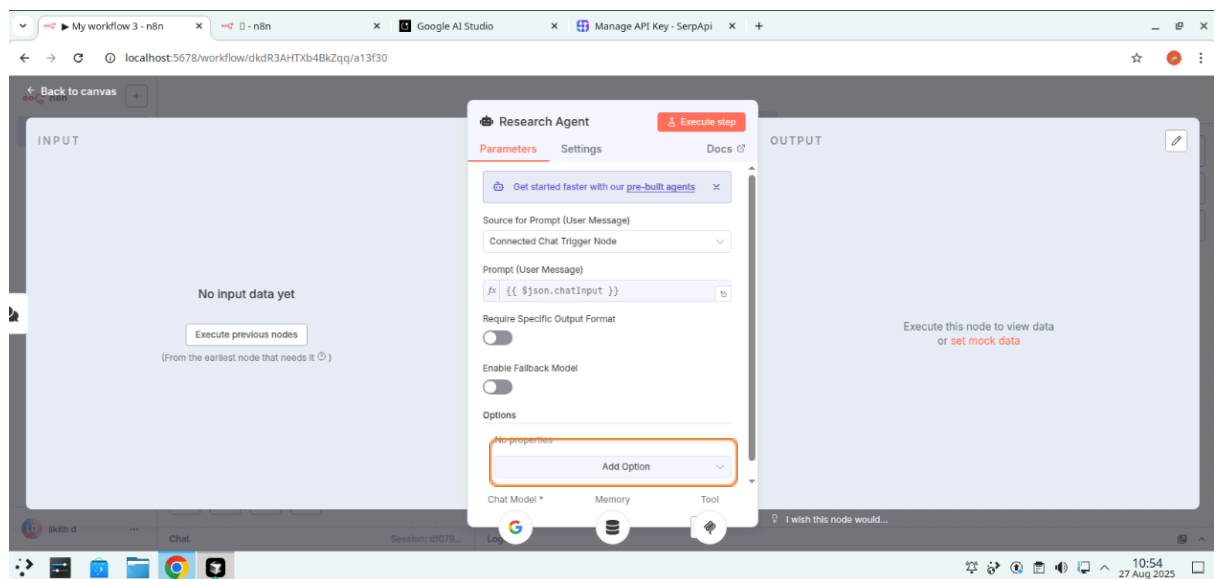


Step 8: Run the Research Agent

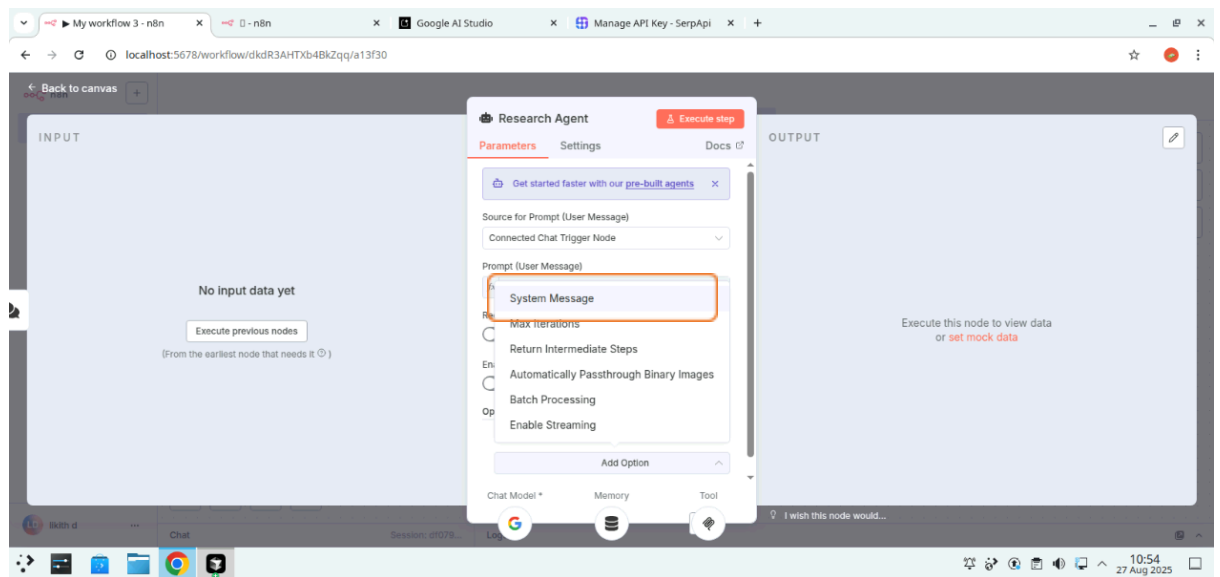
8.1 Return to the **Research Agent** workflow and double-click the **Research Agent** node



8.2 Click the **Add Option** button

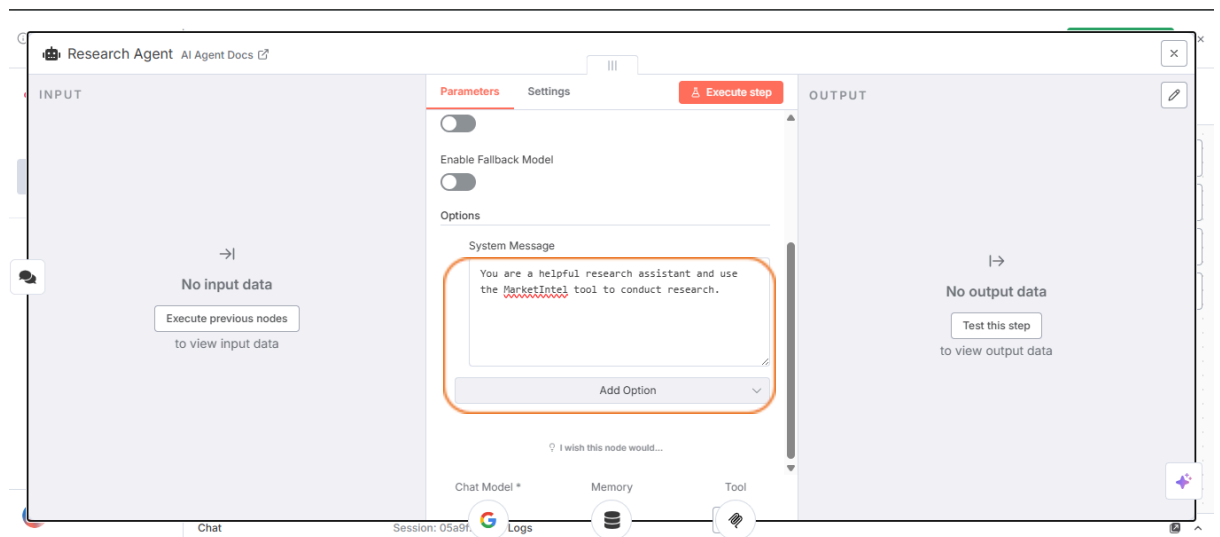


8.3 From the menu, select **System Message**

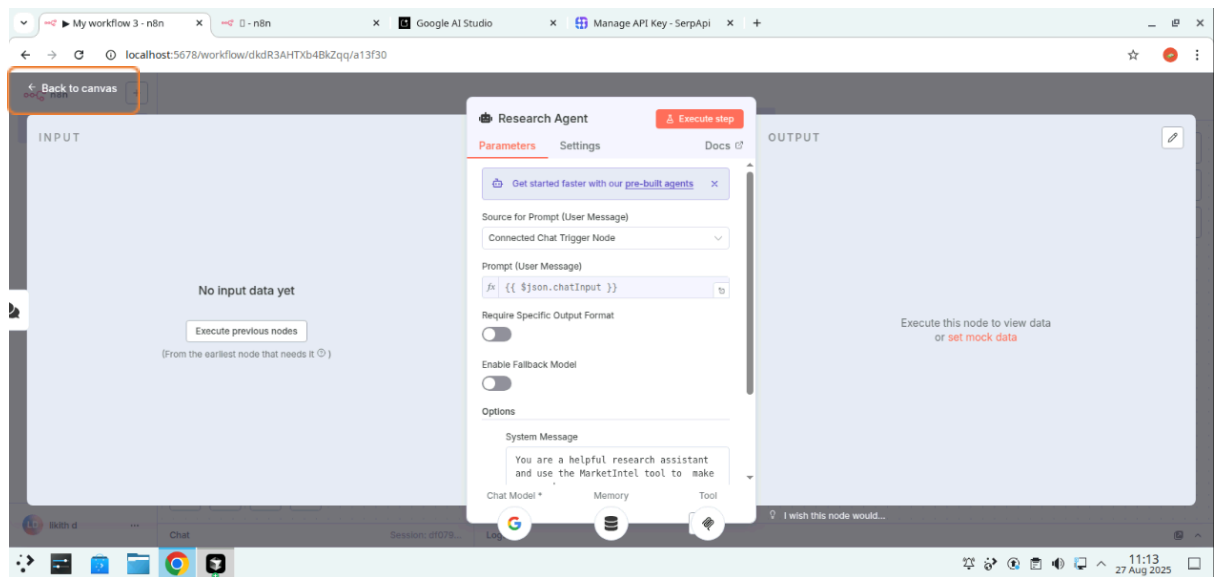


8.4 Delete the default system message and paste the following custom system message:

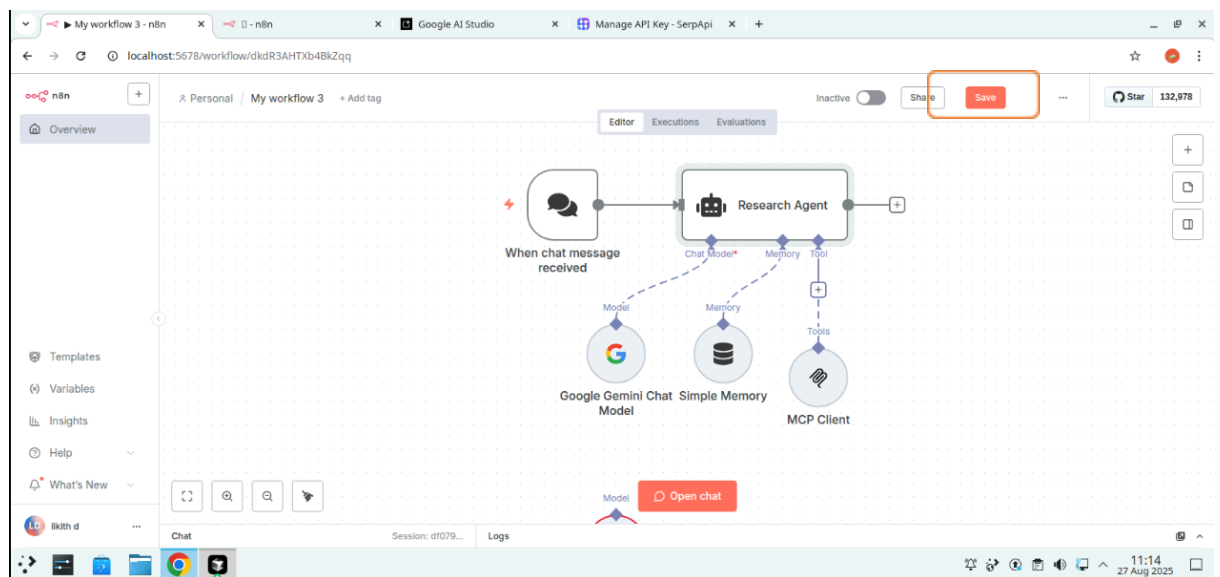
You are a helpful research assistant and use the MarketIntel tool to conduct research.



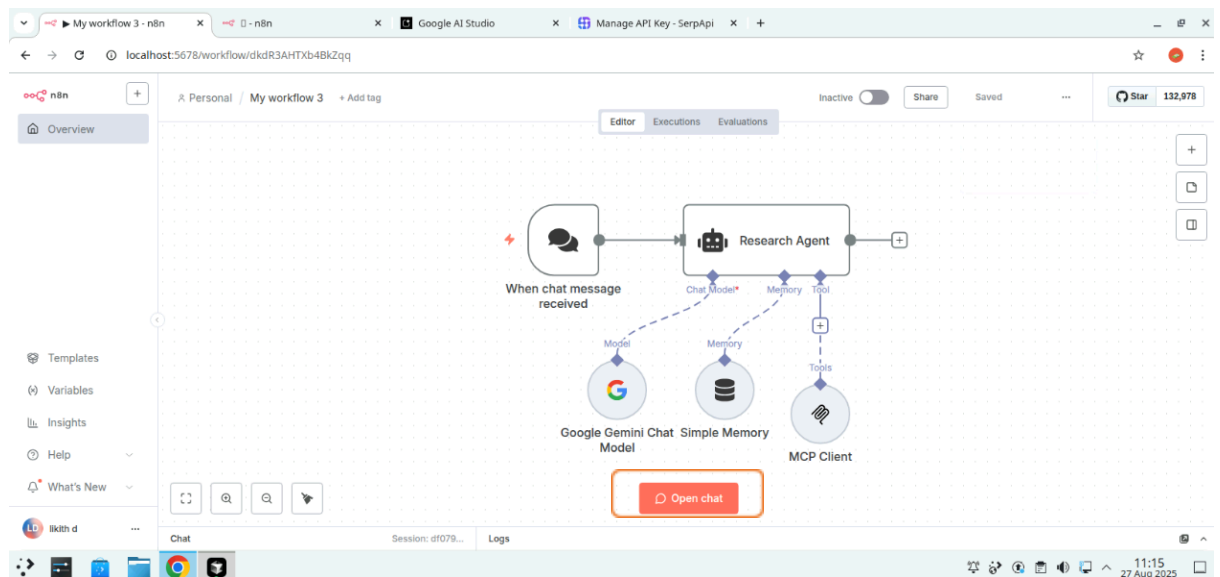
8.5 Click **Back to canvas**



8.6 Save the workflow by clicking the **Save** button in the top right



8.7 Click the **Open Chat** button to launch the chat interface

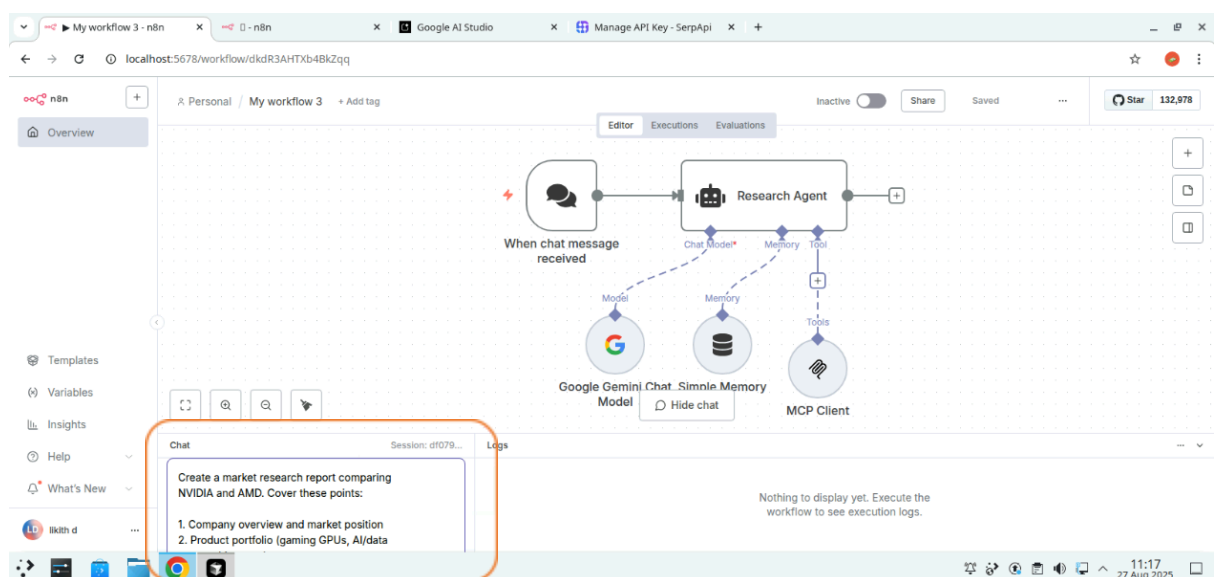


8.8 In the chat window, paste the following test prompt and press **Enter**:

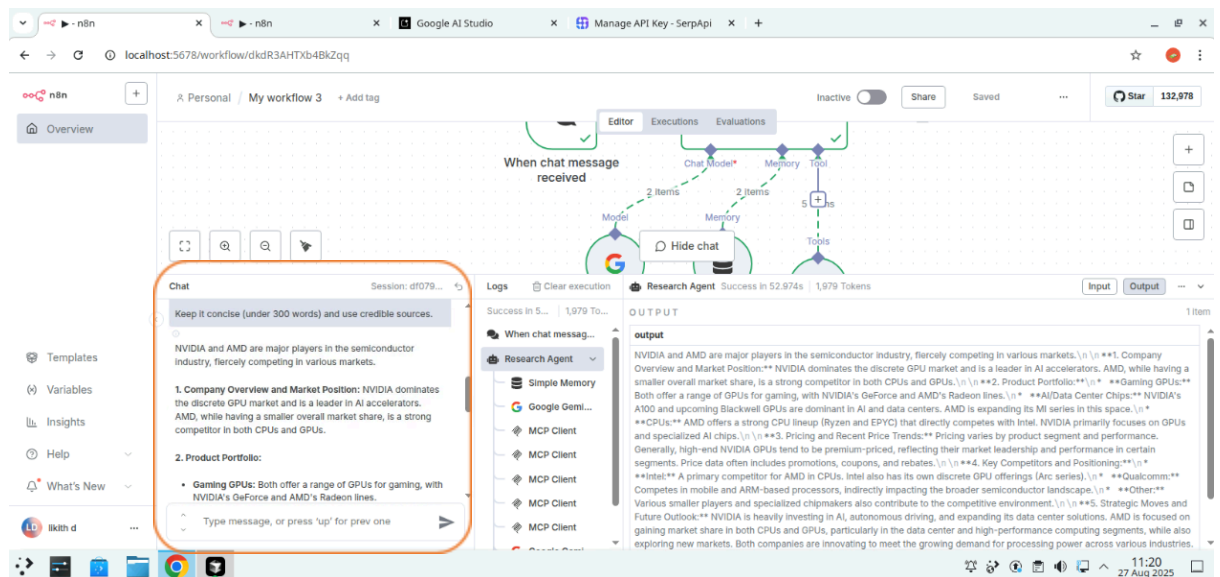
Create a market research report comparing NVIDIA and AMD. Cover these points:

- 1. Company overview and market position**
- 2. Product portfolio (gaming GPUs, AI/data center chips, etc.)**
- 3. Pricing and recent price trends**
- 4. Key competitors (Intel, Qualcomm, etc.) and positioning**
- 5. Strategic moves and future outlook**

Keep it concise (under 300 words) and use credible sources.



The output will appear, showing detailed competitor insights generated by the Research Agent



Note: Before running the Research Agent, make sure that the MCP Server Trigger is activated and the workflow has been executed. Otherwise, the agent will not function properly.

By following these steps, you have successfully integrated the MCP server into an n8n workflow. The setup enables you to connect a Research Agent with chat triggers, memory, MCP tools, and the Google Gemini Chat Model, while also leveraging SerpApi for live data. In the end, you will have a fully functional workflow automation pipeline that generates structured competitor insights directly from chat prompts.